An Arithmetic-Dynamical and Spectral Framework for the Binary Goldbach Conjecture

Dr. Youcef Kelanemer

Ph.D. in Numerical Analysis, Université Paris XI (Paris-Sud)

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Abstract

We introduce an arithmetic-dynamical and spectral framework for the binary Goldbach Conjecture based on the prime divisor operator $D(S) = \bigcup_{p\in S}\{q \in \mathbb{P} \mid q \mid (2N-p)\}$. Under the hypothesis that $2N > 100$ is a counterexample, the iterated sequence $P^*(n+1) = D(P^*(n))$ forms a monotonically non-increasing chain of finite prime sets that stabilizes in finitely many steps at a non-empty stationary fixed point $P^*_\infty = D(P^*_\infty) \subset \mathcal{P}_{\leq \frac{2N-5}{3}}$. Every minimal terminal island $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P^*_\infty$ $(k \geq 2)$ induces a strongly connected directed graph governed by an irreducible exponent matrix $M \in \mathbb{Z}^{k\times k}_{\geq 0}$ with zero diagonal ($\mathrm{Tr}(M) = 0$), row sums $R_i \geq 2$, and spectral radius $\rho(M) \geq 2$.

Combining divisor-gap bounds with Perron–Frobenius theory, we establish universal invariants valid for every minimal terminal island: strict coprimality of the two base complements ($\gcd(2N-p_1, 2N-p_2) = 1$), the sub-square base squeeze ($3 \leq p_1 < p_2 < \sqrt{2N}$ and $0 < \Lambda_{1,2} < \frac{2}{\sqrt{2N}}$), unit in-degree of the maximal prime (in-deg$(p_k) = 1$), and the Perron geometric mean ceiling $\prod_{j=1}^{k} p_j^{u_j} < (2N)^{1/\rho(M)}$. Coupled with Baker–Matveev linear forms in logarithms, polynomial obstructions, and Schur node contraction, these invariants unconditionally rule out $k \in \{0,1\}$, $k = 2$ for $2N \leq 10^{24}$ and for fixed prime bases, square-free systems ($k \in \{3,4\}$, uniformly dense, and symmetric-cofactor islands), and all bounded-height regimes for $k \geq 2$, while eliminating root-compressed high-multiplicity regimes conditionally under the *abc* conjecture and isolating the remaining Diophantine-spectral obstacles.

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1 Introduction

Rather than treating the binary Goldbach problem $2N = p + q$ through exponential sums or combinatorial sieves, this work studies the multiplicative structure of prime complements $2N - p$ under the assumption that an even integer $2N > 100$ is a counterexample. Under this hypothesis, every odd prime $p \in P^*(0) = \{p \in \mathcal{P}_{\leq 2N-3} \mid p \nmid 2N\}$ has a composite odd complement $2N - p$ whose prime factors lie strictly back inside $P^*(0)$. Iterating the prime divisor operator

$$D(S) = \bigcup_{p\in S}\{q \in \mathbb{P} \mid q \mid (2N-p)\}, \qquad P^*(n+1) = D(P^*(n)),$$

therefore generates a finite descending chain $P^*(0) \supsetneq P^*(1) \supseteq \ldots$ that terminates at a non-empty stationary fixed-point set $P^*_\infty = D(P^*_\infty) \subset \mathcal{P}_{\leq \frac{2N-5}{3}}$.

Decomposing P^*_∞ into minimal closed subsets produces minimal terminal islands $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P^*_\infty$ $(k \geq 2)$, each forming a strongly connected digraph governed by an

irreducible exponent matrix $M = (a_{i,j}) \in \mathbb{Z}_{\geq 0}^{k\times k}$ via

$$2N - p_i = \prod_{j=1}^{k} p_j^{a_{i,j}} \quad (i = 1, \ldots, k), \qquad \mathrm{Tr}(M) = 0, \qquad R_i = \sum_{j=1}^{k} a_{i,j} \geq 2.$$

The existence of a Goldbach counterexample is thus equivalent to the solvability of the finite Diophantine-spectral system $\mathcal{S}(2N, k, M)$. Within this framework, we prove:

1. **Universal Arithmetic-Spectral Invariants:** For every minimal island I of cardinality $k \geq 2$, the two smallest primes satisfy $\gcd(2N-p_1, 2N-p_2) = 1$, $3 \leq p_1 < p_2 < \sqrt{2N}$, and $0 < \Lambda_{1,2} = \ln \frac{2N-p_1}{2N-p_2} < \frac{2}{\sqrt{2N}}$ (Theorem 4.3); the maximal prime p_k has incoming degree 1 (Lemma 3.9); and the normalized left Perron eigenvector $\mathbf{u} > \mathbf{0}$ satisfies $\prod_{j=1}^{k} p_j^{u_j} < (2N)^{1/\rho(M)}$ with exact cancellation of $\ln p_k$ whenever $a_{m,k} = 1$ (Theorem 4.4).

2. **Unconditional and Conditional Eliminations:** We unconditionally eliminate $k \in \{0, 1\}$ (Propositions 2.14 and 4.9), $k = 2$ for $2N \leq 10^{24}$, for small exponent pairs $(2, 3), (2, 4), (2, 5)$, and for fixed prime bases via Laurent's two-logarithm theorem (Proposition 4.10), square-free islands for $k \in \{3, 4\}$ as well as uniformly dense and symmetric-cofactor square-free islands for all k (Propositions 4.11 and 4.12), and all bounded-height regimes for $k \geq 2$ via Matveev's theorem (Proposition 4.7). Root-compressed high-multiplicity regimes satisfying $p_2 \prod_{j\in S_1\cup S_2} p_j \leq (2N)^{1-\delta}$ are eliminated conditionally under the *abc* conjecture (Proposition 4.8).

3. **High-Dimensional Contraction and Sub-Square Compression:** Single-parent Schur node contraction accumulates exponent mass into a 5×5 system and bounds $k \leq 97$ at $2N \approx 4 \cdot 10^{18}$ when the contracted average row sum stays below $\lfloor \frac{\ln(2N-3)}{\ln 3} \rfloor$ (Proposition 4.16), while sub-square islands ($p_k \leq \sqrt{2N}, a_{m,k} = 1$) satisfy $R_i \geq 3$ for all i, elevating $\rho(M) \geq 3$ and compressing $\prod_{j=1}^{k} p_j^{u_j} < (2N)^{1/3}$ (Proposition 4.18).

Organization of the Paper. Section 2 defines the divisor operator D and proves convergence to the stationary fixed point P_∞^*. Section 3 establishes the graph topology, exponent matrix representation, Perron–Frobenius spectral identities, and the Triangular Divisor Gap Lemma. Section 4 derives the universal coprime base and spectral invariants and carries out the elimination analysis across dimensions $k \geq 1$. Section 5 summarizes what has been proved, identifies the remaining open obstacles, and outlines potential approaches to resolve them.

2 The Prime Divisor Dynamical System and Convergence to P_∞^*

Let $\mathbb{P}$ denote the set of all prime numbers. Because Goldbach's Conjecture is trivially verified by direct inspection for all even integers $4 \leq 2N \leq 100$, let $2N \in 2\mathbb{N}$ with $2N > 100$ ($N > 50$) be an arbitrary even integer. Throughout this section, we assume that $2N$ is a hypothetical counterexample to Goldbach's Conjecture:

$$\text{Counterexample Hypothesis: } \nexists p_1, p_2 \in \mathbb{P} \text{ such that } 2N = p_1 + p_2.$$

Because $2N > 100$, if a Goldbach partition $2N = p_1 + p_2$ were to exist, both summands would necessarily be odd primes ($p_1, p_2 \geq 3$), since if $p_1 = 2$, then $p_2 = 2N - 2 > 98$ is even and composite.

Proposition 2.1 (Compositeness of the Half-Sum N in a Counterexample)**.** *If $2N > 100$ is a hypothetical counterexample to Goldbach's Conjecture, then N cannot be a prime number; hence N is composite.*

Proof of Proposition 2.1. If N were prime, then $2N = N + N$ would be an explicit representation of $2N$ as the sum of two primes, directly contradicting the hypothesis that $2N$ is a counterexample. Thus N must be composite. □

Definition 2.2 (Initial Search Domain $P^*(0)$)**.** For any even integer $2N > 100$, let $\mathcal{P}_{\leq 2N-3} = \{p \in \mathbb{P} \mid 3 \leq p \leq 2N-3\}$ denote the set of odd primes strictly bounded between 3 and $2N-3$. The initial search domain $P^*(0)$ is defined as the set of primes in $\mathcal{P}_{\leq 2N-3}$ that do not divide $2N$:

$$P^*(0) = \{p \in \mathcal{P}_{\leq 2N-3} \mid p \nmid 2N\}.$$

Remark 2.3 (Exclusion of $2N-1$, 2, and Prime Factors of $2N$)**.** The restriction of the domain $P^*(0)$ to odd primes $3 \leq p \leq 2N-3$ not dividing $2N$ captures all genuine Goldbach candidates and ensures the dynamical validity of the divisor mapping:

1. **Exclusion of $2N-1$:** Even if $2N-1$ is a prime number, it is excluded from $P^*(0)$ for three foundational structural reasons:
   - *No Goldbach Partition:* If $p = 2N-1$, its complement is $2N-(2N-1) = 1$. Since 1 is not a prime number, $2N-1$ can never participate in any Goldbach partition $p+q = 2N$.
   - *Divisor Boundedness:* For every prime $p \in P^*(0)$, since $p \geq 3$, the complement satisfies $2N-p \leq 2N-3 < 2N-1$. Because $0 < 2N-p < 2N-1$, the prime $2N-1$ is strictly too large to divide any complement $2N-p$. Thus, $2N-1$ can never enter the divisor image $D(S)$ for any subset $S \subseteq P^*(0)$.
   - *Operator Well-Definedness:* If $2N-1$ were admitted into the domain, its complement 1 would possess no prime divisors, yielding $D(\{2N-1\}) = \emptyset$, which would artificially degenerate the non-emptiness of the divisor operator. Bounding the domain by $2N-3$ guarantees $2N-p \geq 3 > 1$ for all $p \in P^*(0)$.
2. **Exclusion of $p = 2$:** The complement $2N-2 = 2(N-1) > 98$ is an even integer strictly greater than 2, hence composite.
3. **Exclusion of $p \mid 2N$:** Any prime divisor p of $2N$ necessarily divides its own complement, since $2N-p = p\left(\frac{2N}{p}-1\right)$. Such primes are excluded for two decisive reasons:
   - *No Goldbach Partition:* If the complement $2N-p = q$ were prime, the divisibility $p \mid q$ between primes would force $p = q$, which implies $2N = 2p \implies p = N$. This would require the half-sum N to be prime, which is strictly precluded in a counterexample by Proposition 2.1. Thus, for every prime divisor $p \mid 2N$, the complement $2N-p$ is strictly composite.
   - *Elimination of Trivial Self-Loops ($\mathrm{Tr}(M) = 0$):* Because $p \mid 2N \implies p \mid (2N-p)$, admitting such primes would trivially induce self-loops $p \to p$ in the divisor graph. Restricting the domain to $p \nmid 2N$ ensures that $\gcd(p, 2N-p) = \gcd(p, 2N) = 1$, guaranteeing that no prime can divide its own complement and establishing the zero diagonal ($a_{i,i} = 0$, $\mathrm{Tr}(M) = 0$) of the governing exponent matrix.

Consequently, $P^*(0)$ is both necessary and sufficient for analyzing Goldbach counterexamples.

Definition 2.4 (Prime Divisor Mapping Operator D and Iterated Chains)**.** For any subset of primes $S \subseteq \mathcal{P}_{\leq 2N-3}$, the prime divisor mapping operator $D : \mathcal{P}(\mathcal{P}_{\leq 2N-3}) \to \mathcal{P}(\mathcal{P}_{\leq 2N-3})$ is defined as the union of all prime factors dividing the complements $2N-p$:

$$D(S) = \bigcup_{p \in S} \{q \in \mathbb{P} \mid q \mid (2N-p)\},$$

with $D(\emptyset) = \emptyset$. For a singleton set $S = \{p\}$, we denote $D(\{p\}) = \{q \in \mathbb{P} \mid q \mid (2N - p)\}$. The recursive sequence of iterated divisor sets is given by:

$$P^*(n+1) = D(P^*(n)), \quad \forall n \geq 0.$$

To establish the non-emptiness of $P^*(0)$, we recall the classical form of Bertrand's Postulate (Chebyshev's Theorem):

Theorem 2.5 (Bertrand's Postulate / Chebyshev's Theorem [9])**.** *For any real number $x > 1$, there exists at least one prime number $p \in \mathbb{P}$ satisfying $x < p < 2x$. In particular, for any integer $N > 2$, setting $x = N - 1$ guarantees the existence of at least one prime p in the open interval $(N - 1, 2N - 2)$.*

Proposition 2.6 (Non-Emptiness of the Initial Domain $P^*(0)$)**.** *Under the counterexample hypothesis for $2N > 100$, the initial domain set $P^*(0)$ (Definition 2.2) is non-empty. Specifically,*

$$|P^*(0)| = \pi(2N - 3) - \omega(2N) \geq 1 \implies P^*(0) \neq \emptyset,$$

where $\pi(x)$ is the prime-counting function and $\omega(n)$ is the number of distinct prime factors of n.

Proof of Proposition 2.6. The non-emptiness and cardinality identity of $P^*(0)$ are established as follows:

1. **Exact Cardinality Identity:** The set of all primes $\leq 2N - 3$ has cardinality $\pi(2N - 3)$ and consists of $\{2\} \cup \mathcal{P}_{\leq 2N-3}$. Every prime factor $q \mid 2N$ satisfies either $q = 2$ or $q \mid N$, so $q \leq N < 2N - 3$. Thus all $\omega(2N)$ distinct prime factors of $2N$ (namely $q = 2$ and all odd prime divisors of N) lie in $\{2\} \cup \mathcal{P}_{\leq 2N-3}$. Removing these $\omega(2N)$ prime factors from $\{2\} \cup \mathcal{P}_{\leq 2N-3}$ leaves precisely $P^*(0)$, yielding $|P^*(0)| = \pi(2N - 3) - \omega(2N)$.

2. **Prime Existence via Bertrand's Postulate:** Since $2N > 100 \implies N > 50$, we have $x = N - 1 > 49 > 1$. Applying Theorem 2.5 with $x = N - 1$ guarantees the existence of at least one prime $p \in \mathbb{P}$ satisfying $N - 1 < p < 2(N - 1) = 2N - 2$, which in integers means $N \leq p \leq 2N - 3$.

3. **Exclusion of $p = N$ and Confinement to Candidate Primes:** If $p = N$, then N is prime and $2N = N + N$ is an explicit Goldbach partition; by Proposition 2.1, under the counterexample hypothesis N is composite ($N \notin \mathbb{P}$), so $p \neq N$. Hence p is strictly confined to:
$$N < p \leq 2N - 3.$$

4. **Odd Parity:** Since $p > N > 50$, p is strictly an odd prime, so $p \nmid 2$.

5. **Coprimality with $2N$:** Because $p > N > 0$, p strictly exceeds N and therefore cannot divide N ($p \nmid N$). Combining $p \nmid 2$ and $p \nmid N$ yields:
$$\gcd(p, 2N) = 1 \implies p \nmid 2N.$$

6. **Domain Inclusion & Conclusion:** By Definition 2.2, the conditions $3 \leq p \leq 2N - 3$ and $p \nmid 2N$ place $p \in P^*(0)$. Therefore, $|P^*(0)| \geq 1$ and $P^*(0) \neq \emptyset$.

□

Proposition 2.7 (Absence of Goldbach Partners in a Counterexample)**.** *Let $2N > 100$ be an even integer. Then $2N$ is a counterexample to Goldbach's Conjecture if and only if $N \notin \mathbb{P}$ and for every $p \in P^*(0)$, the complement $2N - p$ is strictly composite. Formally:*

$$2N \text{ is a counterexample} \implies \forall p \in P^*(0), \quad |D(\{p\})| \geq 1 \text{ and } 2N - p \notin \mathbb{P}.$$

Proof of Proposition 2.7. Every integer $m \geq 1$ is either 1, prime, or composite. For any $p \in P^*(0)$, since $p \leq 2N-3$, the complement satisfies $2N-p \geq 3 > 1$, so $2N-p \neq 1$. If $2N$ is a counterexample, $2N-p$ cannot be prime, as $2N = p + (2N-p)$ would be an explicit Goldbach partition into two primes. Therefore, $2N-p$ must be an odd composite integer possessing at least two prime factors (counted with multiplicity) and at least one distinct prime factor, ensuring $|D(\{p\})| \geq 1$. (Conversely, by Remark 2.3, if $N \notin \mathbb{P}$ and $2N-p \notin \mathbb{P}$ for all $p \in P^*(0)$, then no prime $p < 2N$ can sum with another prime to equal $2N$.) □

Proposition 2.8 (Closure of the Divisor Image in the Bounded Domain)**.** *Under the counterexample hypothesis ($2N > 100$), the image of any subset $S \subseteq P^*(0)$ under the divisor mapping operator D (Definition 2.4) is strictly contained within the set of primes less than or equal to $\frac{2N-5}{3}$. Formally:*

$$\forall S \subseteq P^*(0), \quad D(S) \subseteq \mathcal{P}_{\leq \frac{2N-5}{3}} \subset \mathcal{P}_{< \frac{2N}{3}},$$

and in particular, for every singleton element $p \in P^*(0)$, $D(\{p\}) \subseteq \mathcal{P}_{\leq \frac{2N-5}{3}}$.

Proof of Proposition 2.8. The closure and upper bound of the divisor image $D(S)$ are established as follows:

1. **Odd Parity of Domain Elements:** For any $p \in P^*(0)$, $p \geq 3$ is odd by Definition 2.2. Since $2N$ is even, the complement $2N-p$ is strictly odd.

2. **Compositeness of Complements:** By Proposition 2.7, under the counterexample hypothesis $2N-p \geq 3$ is an odd composite integer (≥ 9) with at least two prime factors (counted with multiplicity), all of which are odd primes ≥ 3.

3. **Upper Bound for $p \geq 5$:** If $p \geq 5$, then $2N-p \leq 2N-5$. For any prime factor $q \mid (2N-p)$, the complementary quotient $(2N-p)/q$ is an odd integer ≥ 3. Thus:

$$q \leq \frac{2N-p}{3} \leq \frac{2N-5}{3}.$$

4. **Upper Bound for $p = 3$:** If $p = 3$, since $3 \in P^*(0)$ we have $3 \nmid 2N$, which implies $3 \nmid (2N-3)$. The odd composite $2N-3 > 97$ therefore has only prime factors ≥ 5. Since $2N-3$ is composite, it has at least two prime factors ≥ 5, yielding:

$$q \leq \frac{2N-3}{5} < \frac{2N-5}{3}, \quad \forall 2N > 100.$$

5. **Image Invariance Conclusion:** Every prime factor $q \in D(\{p\})$ is an odd prime satisfying $3 \leq q \leq \frac{2N-5}{3}$. Taking the union over all $p \in S$ guarantees $D(S) \subseteq \mathcal{P}_{\leq \frac{2N-5}{3}} \subset \mathcal{P}_{< \frac{2N}{3}}$.

□

Proposition 2.9 (Non-Divisibility Inheritance (Exclusion of Factors of $2N$))**.** *No prime factor dividing $2N$ can ever enter any iterated set $P^*(n)$ (Definition 2.4). Formally:*

$$\forall q \in \mathbb{P}, \quad q \mid 2N \implies q \notin P^*(n), \quad \forall n \geq 0.$$

Proof of Proposition 2.9. We establish the stronger invariant operator property: for any prime $q \mid 2N$ and any subset $S \subseteq \mathcal{P}_{\leq 2N-3}$,

$$q \in D(S) \implies q \in S.$$

Indeed, if $q \in D(S)$, then by Definition 2.4 there exists some $p \in S$ such that $q \mid (2N-p)$. Because $q \mid 2N$, q divides the linear combination:

$$q \mid (2N - (2N-p)) \implies q \mid p.$$

Since p is prime, its only prime divisor is itself, forcing $q = p \in S$.

In set-theoretic terms, $D(S) \cap \{q \in \mathbb{P} \mid q \mid 2N\} \subseteq S \cap \{q \in \mathbb{P} \mid q \mid 2N\}$. By Definition 2.2, the initial domain $P^*(0)$ contains no prime factor dividing $2N$, so $P^*(0) \cap \{q \in \mathbb{P} \mid q \mid 2N\} = \emptyset$. Iterating D therefore guarantees $P^*(n) \cap \{q \in \mathbb{P} \mid q \mid 2N\} = \emptyset$ for all $n \geq 0$. □

Proposition 2.10 (The Monotonic Nested Chain Property)**.** *The sequence of iterated divisor sets forms a monotonically non-increasing nested chain of finite sets with a strictly proper initial contraction:*

$$P^*(0) \supsetneq P^*(1) \supseteq P^*(2) \supseteq \cdots \supseteq P^*(n) \supseteq P^*(n+1).$$

Proof of Proposition 2.10. The nested sequence structure is proved through the following steps:

1. **Domain Invariance ($D(P^*(0)) \subseteq P^*(0)$):** Let $q \in P^*(1) = D(P^*(0))$. By Proposition 2.8, $q \leq \frac{2N-5}{3} < N \leq 2N-3$. By Proposition 2.9, $q \nmid 2N$. Furthermore, since $2 \mid 2N$, $q \neq 2$, so $q \geq 3$. Hence $q \in \mathcal{P}_{\leq 2N-3}$ and $q \nmid 2N$, which means $q \in P^*(0)$ by Definition 2.2. Therefore, $P^*(1) = D(P^*(0)) \subseteq P^*(0)$.

2. **Strict Initial Contraction ($P^*(0) \supsetneq P^*(1)$):** By Proposition 2.6, $P^*(0)$ contains at least one prime $p \in (N, 2N-3]$. By Proposition 2.8, every element $q \in P^*(1)$ satisfies $q \leq \frac{2N-5}{3} < N$. Thus $p \in P^*(0) \setminus P^*(1)$, proving that $P^*(1)$ is a strictly proper subset of $P^*(0)$:
$$P^*(0) \supsetneq P^*(1).$$

3. **Monotone Chain Induction:** The operator D is monotonic with respect to set inclusion: for any $A \subseteq B \subseteq P^*(0)$,
$$D(A) = \bigcup_{p \in A} D(\{p\}) \subseteq \bigcup_{p \in B} D(\{p\}) = D(B).$$
Since $P^*(1) \subseteq P^*(0)$, applying D yields $P^*(2) = D(P^*(1)) \subseteq D(P^*(0)) = P^*(1)$. Proceeding by induction, $P^*(n+1) = D(P^*(n)) \subseteq D(P^*(n-1)) = P^*(n)$ for all $n \geq 1$.

□

Proposition 2.11 (Convergence to the Stationary Fixed-Point Limit Set P^*_∞)**.** *Because $P^*(0)$ is finite and the sequence $P^*(n)$ undergoes a strictly proper initial contraction followed by a monotonically non-increasing nested chain, there exists a finite iteration index $n_0 \in \mathbb{N}_{\geq 1}$ at which the set stabilizes:*

$$\exists n_0 \in \mathbb{N}_{\geq 1} \quad \textit{such that} \quad \forall n \geq n_0, \quad P^*(n) = P^*(n_0) \equiv P^*_\infty,$$

where P^*_∞ satisfies the stationary identity $D(P^*_\infty) = P^*_\infty$.

Proof of Proposition 2.11. The convergence of the finite nested chain proceeds as follows:

1. **Finiteness of Base Domain:** The initial set $P^*(0)$ is finite with $|P^*(0)| \leq \pi(2N-3) < \infty$.

2. **Non-Increasing Integer Sequence:** By Proposition 2.10, the sequence of cardinalities $(|P^*(n)|)_{n \geq 0}$ is a non-increasing sequence of natural numbers with an initial strict decrease:
$$|P^*(0)| > |P^*(1)| \geq |P^*(2)| \geq \cdots \geq 0.$$

3. **Well-Ordering Stabilization:** By the Well-Ordering Principle of natural numbers, any non-increasing sequence of non-negative integers must stabilize in finitely many steps:
$$\exists n_0 \in \mathbb{N}_{\geq 1} \quad \text{such that} \quad |P^*(n_0)| = |P^*(n_0+1)|.$$

4. **Set Identity & Fixed Point:** Since $P^*(n_0+1) \subseteq P^*(n_0)$ and $|P^*(n_0+1)| = |P^*(n_0)|$, it follows that $P^*(n_0+1) = P^*(n_0) \equiv P_\infty^*$, which satisfies $D(P_\infty^*) = P_\infty^*$. By immediate induction on $n \geq n_0$, $P^*(n) = P_\infty^*$ holds for all $n \geq n_0$.

□

Definition 2.12 (Stationary Fixed-Point Limit Set)**.** A set of primes $S \subseteq \mathcal{P}_{\leq 2N-3}$ is called *stationary* (or a fixed point) under the divisor mapping operator D if $D(S) = S$. The stationary limit set obtained in Proposition 2.11 is denoted $P_\infty^* = D(P_\infty^*)$.

Proposition 2.13 (Non-Emptiness Inheritance of the Divisor Chain)**.** *For any non-empty subset $S \subseteq P^*(0)$, its divisor image $D(S)$ is non-empty. Consequently, under the counterexample hypothesis, every iterated divisor set $P^*(n)$ is non-empty for all $n \geq 0$:*

$$\forall S \subseteq P^*(0), \quad S \neq \emptyset \implies D(S) \neq \emptyset; \quad \textit{hence } P^*(n) \neq \emptyset, \quad \forall n \geq 0.$$

Proof of Proposition 2.13. Non-emptiness propagation follows by induction:

1. **Pointwise Non-Emptiness:** For any $p \in P^*(0)$, $3 \leq p \leq 2N-3 \implies 2N - p \geq 3 > 1$. By the Fundamental Theorem of Arithmetic and Proposition 2.7, $2N - p$ is composite, possessing at least one prime factor q, guaranteeing $|D(\{p\})| \geq 1$ and $D(\{p\}) \neq \emptyset$.
2. **Subset Image Non-Emptiness:** For any non-empty subset $S \subseteq P^*(0)$, selecting $p \in S$ gives $\emptyset \neq D(\{p\}) \subseteq D(S) = \bigcup_{x \in S} D(\{x\})$, forcing $D(S) \neq \emptyset$.
3. **Inductive Base ($n = 0$):** $P^*(0) \neq \emptyset$ by Proposition 2.6.
4. **Inductive Step ($n \to n+1$):** Assuming $P^*(n) \neq \emptyset$, since $P^*(n) \subseteq P^*(0)$ (Proposition 2.10), applying the divisor operator gives $P^*(n+1) = D(P^*(n)) \neq \emptyset$.
5. **Inductive Conclusion:** Therefore, every iterated divisor set satisfies $P^*(n) \neq \emptyset$ for all $n \geq 0$.

□

Proposition 2.14 (Non-Emptiness of the Limit Set P_∞^*)**.** *Under the counterexample assumption, the stationary limit set P_∞^* cannot degenerate to the empty set:*

$$2N \textit{ is a counterexample} \implies P_\infty^* \neq \emptyset \implies |P_\infty^*| \geq 1.$$

Proof of Proposition 2.14. The non-emptiness of P_∞^* is deduced in two steps:

1. **Finite Stabilization Evaluation:** By Proposition 2.11, the sequence of iterated sets stabilizes at a finite index $n_0 \in \mathbb{N}_{\geq 1}$ where $P_\infty^* = P^*(n_0)$.
2. **Inherited Non-Emptiness:** By Proposition 2.13, $P^*(n) \neq \emptyset$ for all $n \geq 0$. Evaluating at $n = n_0$ yields $P_\infty^* = P^*(n_0) \neq \emptyset$, ensuring $|P_\infty^*| \geq 1$.

□

Proposition 2.15 (Structural Ceiling Rule (Compositeness Ceiling))**.** *The maximum prime in P_∞^*, denoted $s = \max(P_\infty^*)$, is strictly bounded above by $\frac{2N-5}{3} < \frac{2N}{3} < N$:*

$$s = \max_{p \in P_\infty^*} p \leq \frac{2N-5}{3} < \frac{2N}{3}.$$

Proof of Proposition 2.15. The ceiling bound is deduced as follows:

1. **Fixed-Point Pre-Image:** Since $P^*_\infty = D(P^*_\infty)$, every prime $p \in P^*_\infty$ is a prime factor of $2N - q$ for some $q \in P^*_\infty \subseteq P^*(0)$.

2. **Divisor Image Ceiling:** By Proposition 2.8, every element of $D(P^*(0))$—and consequently every iterated set $P^*(n)$ for $n \geq 1$ including P^*_∞—is bounded by $\frac{2N-5}{3}$.

3. **Extremal Bound:** Therefore, the maximum prime satisfies $s = \max(P^*_\infty) \leq \frac{2N-5}{3} < \frac{2N}{3} < N$.

□

3 Directed Graph Topology and Exponent Matrix Governance

Having established in Section 2 that a hypothetical counterexample $2N > 100$ forces the existence of a non-empty stationary limit set $P^*_\infty = D(P^*_\infty) \neq \emptyset$ bounded inside $\mathcal{P}_{\leq \frac{2N-5}{3}}$, we now investigate the topological and algebraic structure governing P^*_∞.

Definition 3.1 (Minimal Terminal Islands)**.** A non-empty subset $I \subseteq P^*_\infty$ is called a *minimal terminal island* if $D(I) = I$ and no proper non-empty subset $I' \subsetneq I$ satisfies $D(I') \subseteq I'$.

Proposition 3.2 (Topological Strong Connectivity of Minimal Terminal Islands)**.** *Define the directed relation R on P^*_∞ by $pRq \iff q \mid (2N - p)$. Any minimal terminal island $I \subseteq P^*_\infty$ (Definition 3.1) forms a strongly connected directed graph $G = (I, R)$:*

$$\forall p, q \in I, \quad \exists (v_0, \ldots, v_m) \subseteq I \ (m \geq 1,\ v_0 = p,\ v_m = q) \quad \textit{such that } v_{s-1} R v_s \ \ (\forall 1 \leq s \leq m).$$

Proof of Proposition 3.2. We establish the topological strong connectivity of $G = (I, R)$ by contradiction across four algorithmic steps:

Input / Hypothesis: Suppose for contradiction that the finite directed graph $G = (I, R)$ is not strongly connected. By definition, there exists at least one ordered pair of distinct vertices (p, q) such that q cannot be reached from p via any directed path in G:

$$\exists p, q \in I \quad \text{such that } q \text{ is not reachable from } p \text{ in } G. \tag{1}$$

Step 1 (The Forward Reachable Set Partition): We construct the forward reachable component generated by the source node p:

$$I' := \{p\} \cup \{x \in I \mid \exists \text{ a directed path } p \rightsquigarrow x \text{ in } G\}. \tag{2}$$

This subset satisfies two fundamental properties:

1. *Non-Emptiness:* Since $p \in I'$, the set is strictly non-empty ($I' \neq \emptyset$).

2. *Strict Proper Inclusion:* By hypothesis (1), the target node q is unreachable from p, so $q \notin I'$. Because $q \in I$, this establishes that I' is a strictly proper subset:

$$\emptyset \subsetneq I' \subsetneq I. \tag{3}$$

Step 2 (Forward Invariance under the Divisor Operator D): Let $z \in I'$ be an arbitrary element. We examine the prime divisor image $D(\{z\})$:

1. If $z = p$, every prime factor $y \in D(\{p\})$ satisfies pRy by definition of relation R. Since $D(I) = I$, $y \in I$. Thus (p, y) is a directed edge in G, so y is reachable from p, whence $y \in I'$.

2. If $z \neq p$, z is reachable from p via a directed path $p \rightsquigarrow z$. For any $y \in D(\{z\}) \subseteq D(I) = I$, the relation zRy provides a directed edge (z, y) in G, and concatenating yields a directed path $p \rightsquigarrow y$. Hence y is reachable from p, so $y \in I'$.

Since $D(\{z\}) \subseteq I'$ for every vertex $z \in I'$, taking the union over all elements in I' yields:

$$D(I') = \bigcup_{z \in I'} D(\{z\}) \subseteq I'. \tag{4}$$

Contradiction & Output: Combining (3) and (4) exhibits a non-empty, strictly proper subset $I' \subsetneq I$ satisfying:

$$\emptyset \subsetneq I' \subsetneq I \quad \text{with} \quad D(I') \subseteq I'. \tag{5}$$

Relation (5) directly violates the defining minimality of the terminal island I (Definition 3.1), which requires that no non-empty proper subset can be closed under D.

This contradiction refutes (1), establishing unconditionally that $G = (I, R)$ is topologically strongly connected.

This strong connectivity is vital for the entire subsequent framework: it proves that a minimal terminal island contains no transient feeder nodes or subordinate basins of attraction. Every prime $p_i \in I$ communicates cyclically with every other prime $p_j \in I$. □

Proposition 3.3 (Existence and Cardinality Floor of Minimal Terminal Islands)**.** *Under the counterexample hypothesis, the stationary limit set P^*_∞ contains at least one minimal terminal island $I \subseteq P^*_\infty$ satisfying $D(I) = I$. Every such minimal island I has cardinality $k = |I| \geq 2$ and forms a strongly connected digraph $G = (I, R)$:*

$$\exists I \subseteq P^*_\infty \quad \textit{such that} \quad D(I) = I, \quad G = (I, R) \textit{ is strongly connected, and } k = |I| \geq 2.$$

Proof of Proposition 3.3. The existence and cardinality floor of a minimal terminal island are established as follows:

1. **Family of Invariant Subsets:** Define $\mathcal{F} = \{S \subseteq P^*_\infty \mid S \neq \emptyset \text{ and } D(S) \subseteq S\}$. By Proposition 2.14, $P^*_\infty \neq \emptyset$, and by Proposition 2.11, $D(P^*_\infty) = P^*_\infty \subseteq P^*_\infty$. Hence $P^*_\infty \in \mathcal{F}$, so $\mathcal{F}$ is non-empty.

2. **Existence of a Minimal Element:** Because P^*_∞ is finite, $\mathcal{F}$ is a non-empty family of finite sets. Under the partial order of set inclusion ($\subseteq$), every non-empty finite family of sets possesses at least one minimal element $I \in \mathcal{F}$.

3. **Stationary Identity ($D(I) = I$):** By definition of $\mathcal{F}$, $D(I) \subseteq I$. Since $I \neq \emptyset$, Proposition 2.13 guarantees that $D(I) \neq \emptyset$. Monotonicity of D yields $D(D(I)) \subseteq D(I)$, which implies $D(I) \in \mathcal{F}$. By set minimality of $I \in \mathcal{F}$, the inclusion $D(I) \subseteq I$ forces $D(I) = I$.

4. **Strict Minimality:** If any proper non-empty subset $I' \subsetneq I$ satisfied $D(I') \subseteq I'$, then I' would belong to $\mathcal{F}$, contradicting the minimality of I. Thus I satisfies Definition 3.1.

5. **Cardinality Floor ($k \geq 2$):** If $|I| = 1$, say $I = \{p\}$, then $D(I) = I$ would imply $D(\{p\}) = \{p\}$, meaning $p \mid (2N - p)$, which forces $p \mid 2N$. However, by Non-Divisibility Inheritance (Proposition 2.9), no prime in $P^*(0)$ divides $2N$. Since $I \subseteq P^*_\infty \subseteq P^*(0)$, this is an immediate contradiction. Thus every invariant subset $S \in \mathcal{F}$ satisfies $|S| \geq 2$, forcing $k = |I| \geq 2$.

6. **Strong Connectivity:** By Proposition 3.2, $G = (I, R)$ is strongly connected.

□

Remark 3.4 (Structure of P_∞^* and Sufficiency of Terminal Island Elimination)**.** The stationary limit set P_∞^* need not be a disjoint union of minimal terminal islands; it may in principle contain transient primes that flow into a minimal terminal island under iterated applications of D without being reachable from it. However, because every non-empty stationary set $P_\infty^* \neq \emptyset$ is mathematically guaranteed to contain at least one minimal terminal island $I \subseteq P_\infty^*$ (Proposition 3.3), establishing the complete non-existence of minimal terminal islands across all cardinalities $k \geq 2$ unconditionally forces $P_\infty^* = \emptyset$. This rigorously reduces the global elimination of counterexamples entirely to the elimination of minimal terminal islands.

To formalize the equivalence between directed graph connectivity and matrix irreducibility, we state the classical theorem of Frobenius and Varga:

Theorem 3.5 (Frobenius–Varga Theorem on Matrix Irreducibility [4, 10])**.** *Let $M = (a_{i,j}) \in \mathbb{R}_{\geq 0}^{k \times k}$ be a non-negative $k \times k$ matrix ($k \geq 2$), and let $G(M) = (V, E)$ be its associated directed graph with vertex set $V = \{1, \ldots, k\}$ and directed edges $(i, j) \in E \iff a_{i,j} > 0$. Then the following statements are equivalent:*

1. ***Algebraic Irreducibility:*** *The matrix M is irreducible; that is, there exists no permutation matrix P such that*
$$P^T M P = \begin{pmatrix} M_{1,1} & M_{1,2} \\ \mathbf{0} & M_{2,2} \end{pmatrix},$$
where $M_{1,1}$ and $M_{2,2}$ are square submatrices of dimensions $m \times m$ and $(k-m) \times (k-m)$ respectively ($1 \leq m < k$).

2. ***Topological Strong Connectivity:*** *The associated directed graph $G(M)$ is strongly connected; that is, for every pair of distinct vertices $i, j \in V$, there exists a directed path from i to j in $G(M)$.*

Proposition 3.6 (Exponent Matrix Representation and Irreducibility)**.** *Let $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P_\infty^*$ be a minimal terminal island ($k \geq 2$) ordered in strictly ascending order. Then I is uniquely governed by a non-negative integer exponent matrix $M = (a_{i,j}) \in \mathbb{Z}_{\geq 0}^{k \times k}$ with zero diagonal ($\mathrm{Tr}(M) = 0$):*

$$\forall i \in \{1, \ldots, k\}, \quad 2N - p_i = \prod_{j=1}^{k} p_j^{a_{i,j}}, \quad \text{with } a_{i,i} = 0.$$

Moreover, the matrix M is algebraically irreducible.

Proof of Proposition 3.6. The representation and irreducibility are established as follows:

1. **Prime Factorization in I:** Since $D(I) = I$, the set of prime factors of $2N - p_i$ is contained in I for every $p_i \in I$. By the Fundamental Theorem of Arithmetic and the fixed ascending ordering $p_1 < p_2 < \cdots < p_k$, each complement $2N - p_i$ factors uniquely as $2N - p_i = \prod_{j=1}^{k} p_j^{a_{i,j}}$ for integers $a_{i,j} \geq 0$.

2. **Zero Diagonal ($\mathrm{Tr}(M) = 0$):** By Non-Divisibility Inheritance (Proposition 2.9), no prime $p_i \in I \subseteq P^*(0)$ divides $2N$. If $p_i \mid (2N - p_i)$, then $p_i \mid 2N$, a contradiction. Thus $p_i \nmid (2N - p_i)$, which enforces $a_{i,i} = 0$ for all $i \in \{1, \ldots, k\}$, so $\mathrm{Tr}(M) = \sum_{i=1}^{k} a_{i,i} = 0$.

3. **Graph-Matrix Equivalence & Irreducibility:** The directed graph $G(M)$ of the non-negative matrix M has a directed edge from i to j if and only if $a_{i,j} \geq 1$, which is equivalent to $p_j \mid (2N - p_i)$, matching the relation R of $G = (I, R)$. By Proposition 3.2, $G = (I, R)$ is strongly connected. By the Frobenius–Varga Theorem on Matrix Irreducibility (Theorem 3.5), because $G(M)$ is strongly connected and $k \geq 2$, the matrix M is algebraically irreducible.

□

To establish the spectral properties of the governing exponent matrix, we state the classical spectral theorem of Perron, Frobenius, Collatz, and Wielandt:

Theorem 3.7 (Perron–Frobenius & Collatz–Wielandt Spectral Theorem [3,4,8,11])**.** *Let $M = (a_{i,j}) \in \mathbb{R}_{\geq 0}^{k \times k}$ be a non-negative, algebraically irreducible matrix of dimension $k \geq 2$, and let $\rho(M)$ denote its spectral radius. Then:*

1. ***Simplicity and Positivity of the Spectral Radius:*** *The spectral radius $\rho(M) > 0$ is a simple positive real eigenvalue of M, and every eigenvalue λ of M satisfies $|\lambda| \leq \rho(M)$.*

2. ***Strictly Positive Left and Right Eigenvectors:*** *There exists a unique normalized left eigenvector $\mathbf{u} = (u_1, \ldots, u_k)^T > \mathbf{0}$ and a unique normalized right eigenvector $\mathbf{v} = (v_1, \ldots, v_k)^T > \mathbf{0}$ satisfying:*

$$\mathbf{u}^T M = \rho(M)\mathbf{u}^T, \quad M\mathbf{v} = \rho(M)\mathbf{v},$$

   where $u_i > 0$ and $v_i > 0$ for all $i \in \{1, \ldots, k\}$, normalized such that $\sum_{i=1}^{k} u_i = 1$.

3. ***Collatz–Wielandt Row-Sum Bounds:*** *The spectral radius $\rho(M)$ is bounded between the minimum and maximum row sums of M:*

$$\min_{1 \leq i \leq k} \sum_{j=1}^{k} a_{i,j} \leq \rho(M) \leq \max_{1 \leq i \leq k} \sum_{j=1}^{k} a_{i,j}. \tag{6}$$

   In particular, if every row sum satisfies $\sum_{j=1}^{k} a_{i,j} \geq 2$, then $\rho(M) \geq 2$.

Proposition 3.8 (Linear Logarithmic System, Row Sums, and Spectral Radius Lower Bound)**.** *Let $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P_\infty^*$ be a minimal terminal island ($k \geq 2$), governed by the irreducible exponent matrix $M \in \mathbb{Z}_{\geq 0}^{k \times k}$. Then:*

1. ***Minimum Row Sum:*** *Every row sum of M satisfies:*

$$\sum_{j=1}^{k} a_{i,j} \geq 2, \quad \forall i \in \{1, \ldots, k\}.$$

2. ***Linear Logarithmic System:*** *Defining the logarithmic vectors $\mathbf{x} = (\ln p_1, \ldots, \ln p_k)^T \in \mathbb{R}_{>0}^k$ and $\mathbf{y} = (\ln(2N - p_1), \ldots, \ln(2N - p_k))^T \in \mathbb{R}_{>0}^k$, the factorization system is equivalent to the matrix equation:*

$$\mathbf{y} = M\mathbf{x}.$$

3. ***Perron–Frobenius Spectral Identity:*** *By Theorem 3.7, there exists a unique strictly positive left Perron eigenvector $\mathbf{u} = (u_1, \ldots, u_k)^T > \mathbf{0}$ with $\sum_{i=1}^{k} u_i = 1$ such that $\mathbf{u}^T M = \rho(M)\mathbf{u}^T$, yielding the exact spectral quotient:*

$$\rho(M) = \frac{\mathbf{u}^T \mathbf{y}}{\mathbf{u}^T \mathbf{x}} = \frac{\sum_{i=1}^{k} u_i \ln(2N - p_i)}{\sum_{j=1}^{k} u_j \ln p_j}.$$

4. ***Spectral Radius Lower Bound:*** *By Theorem 3.7 (Collatz–Wielandt bounds (6)), the spectral radius satisfies:*

$$\rho(M) \geq \min_{1\leq i\leq k} \sum_{j=1}^{k} a_{i,j} \geq 2.$$

5. ***Root Compression and Exponent Bounds:*** *For all $i, j \in \{1, \ldots, k\}$,*

$$a_{i,j} \leq \left\lfloor \frac{\ln(2N-3)}{\ln 3} \right\rfloor, \quad \text{and} \quad (a_{i,j} \geq 2 \implies p_j \leq \sqrt{2N-3}).$$

Proof of Proposition 3.8. We establish the five properties in sequence:

1. **Row Sum Bound:** For every $p_i \in I$, $2N - p_i$ is composite by Proposition 2.7. Since $p_i \geq 3$ and $2N$ is even, $2N - p_i$ is an odd composite integer, possessing at least two prime factors (counted with multiplicity), all of which are ≥ 3. Because $D(I) = I$, all prime factors belong to I, so the row sum $\sum_{j=1}^{k} a_{i,j}$ counts the total number of prime factors of $2N - p_i$. Hence $\sum_{j=1}^{k} a_{i,j} \geq 2$ for every $i \in \{1, \ldots, k\}$.

2. **Linear Logarithmic System:** Taking natural logarithms of $2N - p_i = \prod_{j=1}^{k} p_j^{a_{i,j}}$ gives:

$$\ln(2N - p_i) = \sum_{j=1}^{k} a_{i,j} \ln p_j, \quad \forall i \in \{1, \ldots, k\},$$

   which in vector form is identically $\mathbf{y} = M\mathbf{x}$.

3. **Perron–Frobenius Spectral Identity:** Because $M \in \mathbb{Z}_{\geq 0}^{k\times k}$ is non-negative and irreducible with $k \geq 2$ (Proposition 3.6), Theorem 3.7 (Perron–Frobenius) ensures that the spectral radius $\rho(M)$ is a simple, strictly positive eigenvalue, with a unique normalized left eigenvector $\mathbf{u} = (u_1, \ldots, u_k)^T$ satisfying $\mathbf{u}^T M = \rho(M)\mathbf{u}^T$ with $u_i > 0$ for all i and $\sum_{i=1}^{k} u_i = 1$. Multiplying $\mathbf{y} = M\mathbf{x}$ from the left by $\mathbf{u}^T$ yields:

$$\mathbf{u}^T\mathbf{y} = \mathbf{u}^T(M\mathbf{x}) = (\mathbf{u}^T M)\mathbf{x} = \rho(M)\mathbf{u}^T\mathbf{x}.$$

   Since $p_j \geq 3 > 1$, we have $\ln p_j > 0$. Combined with $u_j > 0$, the denominator $\mathbf{u}^T\mathbf{x} = \sum_{j=1}^{k} u_j \ln p_j > 0$ is strictly positive, giving the quotient identity $\rho(M) = \frac{\mathbf{u}^T\mathbf{y}}{\mathbf{u}^T\mathbf{x}}$.

4. **Spectral Radius Lower Bound:** By the Collatz–Wielandt row-sum bounds in Theorem 3.7 (6), the spectral radius $\rho(M)$ is bounded below by the minimum row sum. Directly from the left Perron identity $\mathbf{u}^T M = \rho(M)\mathbf{u}^T$ and the all-ones vector $\mathbf{1} = (1, \ldots, 1)^T$:

$$\rho(M) = \mathbf{u}^T M\mathbf{1} = \sum_{i=1}^{k} u_i \left(\sum_{j=1}^{k} a_{i,j} \right) \geq \min_{1\leq i\leq k} \sum_{j=1}^{k} a_{i,j} \geq 2.$$

5. **Root Compression and Exponent Bounds:** For any entry $a_{i,j} \geq 1$, since $p_j \geq 3$ and $p_i \geq 3$, we have:

$$3^{a_{i,j}} \leq p_j^{a_{i,j}} \leq 2N - p_i \leq 2N - 3 \implies a_{i,j} \leq \frac{\ln(2N-3)}{\ln 3}.$$

   Furthermore, if $a_{i,j} \geq 2$, then $p_j^2 \leq p_j^{a_{i,j}} \leq 2N - p_i \leq 2N - 3$, forcing $p_j \leq \sqrt{2N-3}$.

□

Lemma 3.9 (The Triangular Divisor Gap Lemma)**.** *Let $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P_\infty^*$ be a minimal terminal island ($k \geq 2$). For any two primes $p_a, p_b \in I$ with $p_a < p_b$, if an odd prime $q \in \mathbb{P}$ divides both complements $2N - p_a$ and $2N - p_b$, then:*

$$q \leq \frac{p_b - p_a}{2} < p_b.$$

In particular, no prime $q \geq p_b$ can ever divide both $2N - p_a$ and $2N - p_b$. Consequently:

1. ***Coprime Base Complements:*** *The complements of the two smallest primes share no prime factors from the island:*

$$\gcd(2N - p_1, 2N - p_2) = 1 \implies a_{1,j}a_{2,j} = 0 \quad (\forall j \in \{1, \ldots, k\}).$$

2. ***Universal In-Degree Rigidity of the Maximal Prime:*** *The maximal prime $p_k = \max(I)$ can divide at most one complement $2N - p_i$ ($i < k$). Combined with strong connectivity (*in-deg$(p_k) \geq 1$*), p_k has* ***in-degree exactly*** *1: there is a unique incoming node p_m ($m < k$) such that $a_{m,k} \geq 1$, while $a_{i,k} = 0$ for all $i \neq m$.*

3. ***Sub-Maximal In-Degree Rigidity:*** *For every index $r \in \{2, \ldots, k\}$, the prime p_r can divide at most one complement among $\{2N - p_1, \ldots, 2N - p_{r-1}\}$.*

Proof of Lemma 3.9. If $q \mid (2N - p_a)$ and $q \mid (2N - p_b)$, then q divides their positive difference:

$$(2N - p_a) - (2N - p_b) = p_b - p_a > 0.$$

Since $p_a < p_b$ are both odd primes (≥ 3), their difference $p_b - p_a$ is an even positive integer, while $q \geq 3$ is odd, so $q \mid \frac{p_b - p_a}{2}$. This forces $q \leq \frac{p_b - p_a}{2} < p_b$. If $q \geq p_b$, we obtain $p_b \leq q \leq \frac{p_b - p_a}{2} < p_b$, an immediate contradiction.

Properties (1), (2), and (3) follow directly:

1. In a minimal terminal island, all prime factors of $2N - p_1$ and $2N - p_2$ belong to I. Since $a_{1,1} = 0$ and $a_{2,2} = 0$, any shared factor $q \in I$ must satisfy $q \notin \{p_1, p_2\}$, hence $q \geq p_3 > p_2$. But the gap inequality forces $q < p_2$, an immediate contradiction. Thus $\gcd(2N - p_1, 2N - p_2) = 1$.

2. For any pair $p_a < p_b < p_k$, we have $p_b < p_k$, so p_k cannot divide both $2N - p_a$ and $2N - p_b$. Hence p_k can divide at most one complement from $\{p_1, \ldots, p_{k-1}\}$. Because $G = (I, R)$ is strongly connected (Proposition 3.2), in-deg$(p_k) \geq 1$, which forces in-deg$(p_k) = 1$.

3. The identical argument applies to any pair $p_a < p_b < p_r$, showing that p_r cannot divide both $2N - p_a$ and $2N - p_b$.

□

The Governing Linear-Algebraic & Diophantine System $\mathcal{S}(2N, k, M)$

The algebraic, spectral, and Diophantine constraints governing any hypothetical minimal terminal island in the matrix system $\mathcal{S}(2N, k, M)$ (where $M \in \mathbb{Z}_{\geq 0}^{k \times k}$, $k \geq 2$, is an **irreducible non-negative integer matrix** with $\mathrm{Tr}(M) = 0$) are summarized by the 8-condition system:

$$\begin{cases} 1.\textbf{Prime Domain Bounds:} & 3 \le p_1 < p_2 < \cdots < p_k \le \frac{2N-5}{3}, \quad p_i \in \mathbb{P}, \quad k \ge 2 \\ 2.\textbf{Diophantine Factorization:} & 2N - p_i = \prod_{j=1}^{k} p_j^{a_{i,j}}, \quad \forall i \in \{1, \ldots, k\} \\ 3.\textbf{Zero Diagonal Identity:} & a_{i,i} = 0, \quad \forall i \in \{1, \ldots, k\} \quad (\mathrm{Tr}(M) = 0) \\ 4.\textbf{Minimum Row Sum:} & \sum_{j=1}^{k} a_{i,j} \ge 2, \quad \forall i \in \{1, \ldots, k\} \\ 5.\textbf{Spectral Radius Bound:} & \rho(M) = \frac{\mathbf{u}^T \mathbf{y}}{\mathbf{u}^T \mathbf{x}} = \frac{\sum u_i \ln(2N - p_i)}{\sum u_j \ln p_j} \ge 2 \quad (\mathbf{u} > \mathbf{0}, \sum u_i = 1) \\ 6.\textbf{Exponent Lattice Ceiling:} & a_{i,j} \in \left\{0, 1, \ldots, \left\lfloor \frac{\ln(2N-3)}{\ln 3} \right\rfloor \right\} \\ 7.\textbf{Root Compression Constraint:} & (a_{i,j} \ge 2) \implies p_j \le \sqrt{2N-3} \\ 8.\textbf{Coprime Base \& In-Degree Rigidity:} & \gcd(2N - p_1, 2N - p_2) = 1 \ (a_{1,j} a_{2,j} = 0), \quad \text{in-deg}(p_k) = 1 \end{cases}$$

4 Unified Structural Invariants and Low-Dimensional Elimination

Rather than analyzing each cardinality $k = |I|$ in isolation, we establish a unified, dimension-independent suite of arithmetic, spectral, and Diophantine laws that hold simultaneously for **every** minimal terminal island $I = \{p_1 < \cdots < p_k\} \subseteq P_\infty^*$ $(k \ge 2)$. Specializing these universal invariants yields direct eliminations for low-dimensional topologies $(k = 1, 2, 3, 4)$ and sharply constrains all higher-dimensional configurations.

4.1 Universal Structural Laws for All Minimal Terminal Islands ($k \ge 2$)

Proposition 4.1 (Uniform Logarithmic Exponent Bound)**.** *Assuming $2N > 100$ is a Goldbach counterexample, the entries of the exponent matrix $M = (a_{i,j})_{k \times k}$ governing any minimal terminal island $I \subseteq P_\infty^*$ $(k \ge 2)$ are uniformly bounded by:*

$$\max_{1 \le i,j \le k} a_{i,j} \le \left\lfloor \frac{\ln(2N-3)}{\ln 3} \right\rfloor = O(\ln N).$$

In particular, at the empirical verification benchmark $2N \approx 4 \cdot 10^{18}$ [7], every exponent satisfies $a_{i,j} \le 38 < 40$.

Proof of Proposition 4.1. For any prime $p_i, p_j \in I \subset P^*(0)$, since $p_i, p_j \ge 3$ are odd primes, $2N - p_i \le 2N - 3$. Because $2N - p_i = \prod_{m=1}^{k} p_m^{a_{i,m}}$ with all $p_m \ge 3$, each individual prime power satisfies $3^{a_{i,j}} \le p_j^{a_{i,j}} \le 2N - p_i \le 2N - 3$. Taking natural logarithms yields $a_{i,j} \le \lfloor \frac{\ln(2N-3)}{\ln 3} \rfloor$. □

Proposition 4.2 (Exponent Root-Compression Bound)**.** *Let $I = \{p_1 < \cdots < p_k\} \subseteq P_\infty^*$ $(k \ge 2)$ be a minimal terminal island. If any exponent in M satisfies $a_{i,j} \ge 2$, then the repeated prime p_j is strictly confined below $\sqrt{2N-3}$, and any distinct co-factor prime $p_r \in I$ dividing $2N - p_i$ $(r \ne j)$ is bounded by $\frac{2N-3}{9}$:*

$$(a_{i,j} \ge 2) \implies p_j \le \sqrt{2N-3} < \sqrt{2N} \quad \textit{and} \quad (a_{i,r} \ge 1, r \ne j) \implies p_r \le \frac{2N-3}{9}.$$

Proof of Proposition 4.2. Suppose $a_{i,j} \ge 2$ for some $i, j \in \{1, \ldots, k\}$. Since $p_i \ge 3$:

$$p_j^2 \le p_j^{a_{i,j}} \prod_{m \ne j} p_m^{a_{i,m}} = 2N - p_i \le 2N - 3 \implies p_j \le \sqrt{2N-3} < \sqrt{2N}. \tag{7}$$

If $a_{i,r} \ge 1$ for some $r \ne j$, then since $p_j \ge 3 \implies p_j^2 \ge 9$, we have $9p_r \le p_j^2 p_r \le 2N - p_i \le 2N - 3 \implies p_r \le \frac{2N-3}{9}$. □

We now establish the first universal structural theorem governing the base primes $p_1 < p_2$ across **all** dimensions $k \geq 2$. Throughout the remainder of the paper, for each row $i \in \{1, \ldots, k\}$, we denote its **index support** by $S_i := \{j \in \{1, \ldots, k\} \mid a_{i,j} \geq 1\}$ and its **row sum** (total prime factor multiplicity of $2N - p_i$) by $R_i := \sum_{j=1}^{k} a_{i,j} = \Omega(2N - p_i)$.

Theorem 4.3 (The Universal Coprime Base & Sub-Square Threshold Theorem)**.** *Let $2N > 100$ be a hypothetical Goldbach counterexample, and let $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P_\infty^*$ be a minimal terminal island of **any** cardinality $k \geq 2$. Then:*

1. ***Universal Sub-Square Base Ceiling** ($p_1 < p_2 < \sqrt{2N}$)**:** The two smallest primes in I unconditionally satisfy:*

$$3 \leq p_1 < p_2 \leq \sqrt{2N - p_1} \leq \sqrt{2N - 3} < \sqrt{2N}. \tag{8}$$

2. ***Strict Coprimality of Base Complements:** The two largest complements $2N - p_1$ and $2N - p_2$ are strictly coprime, with completely disjoint prime supports:*

$$\gcd(2N - p_1, 2N - p_2) = 1 \iff S_1 \cap S_2 = \emptyset, \tag{9}$$

where $S_1 \subseteq \{2, 3, \ldots, k\}$ and $S_2 \subseteq \{1, 3, \ldots, k\}$.

3. ***Universal Coprime Proximity Identity:** The disjoint prime factorizations of $2N - p_1$ and $2N - p_2$ satisfy the exact additive-multiplicative bridge:*

$$\prod_{j \in S_1} p_j^{a_{1,j}} - \prod_{r \in S_2} p_r^{a_{2,r}} = p_2 - p_1 < \sqrt{2N - 3}, \tag{10}$$

and their logarithmic linear form $\Lambda_{1,2} := \ln\left(\frac{2N-p_1}{2N-p_2}\right) = \sum_{j \in S_1} a_{1,j} \ln p_j - \sum_{r \in S_2} a_{2,r} \ln p_r$ satisfies the strict proximity sandwich:

$$0 < \Lambda_{1,2} < \frac{p_2 - p_1}{2N - p_2} < \frac{\sqrt{2N}}{2N - \sqrt{2N}} < \frac{2}{\sqrt{2N}}. \tag{11}$$

In particular, for all $2N > 4 \cdot 10^{18}$, $0 < \Lambda_{1,2} < 10^{-9}$.

Proof of Theorem 4.3. 1. **Proof of (1):** Consider the first row $2N - p_1 = \prod_{j=1}^{k} p_j^{a_{1,j}}$. By the zero-diagonal property (Proposition 3.6), $a_{1,1} = 0$, so every prime factor of $2N - p_1$ belongs to $\{p_2, p_3, \ldots, p_k\}$ and therefore satisfies $p_j \geq p_2$. By Proposition 3.8, the row sum satisfies $R_1 = \sum_{j=2}^{k} a_{1,j} \geq 2$. Consequently:

$$2N - 3 \geq 2N - p_1 = \prod_{j=2}^{k} p_j^{a_{1,j}} \geq p_2^{\sum_{j=2}^{k} a_{1,j}} \geq p_2^2 \implies p_2 \leq \sqrt{2N - p_1} \leq \sqrt{2N - 3} < \sqrt{2N}.$$

2. **Proof of (2):** This follows directly from Lemma 3.9 (Part 1): any prime $q \in I$ dividing both $2N - p_1$ and $2N - p_2$ must divide their difference $p_2 - p_1$, forcing $q \leq \frac{p_2 - p_1}{2} < p_2$, whereas $a_{1,1} = a_{2,2} = 0 \implies q \geq p_3 > p_2$, a contradiction.

3. **Proof of (3):** Subtracting $2N - p_2 = \prod_{r \in S_2} p_r^{a_{2,r}}$ from $2N - p_1 = \prod_{j \in S_1} p_j^{a_{1,j}}$ yields (10), and $p_2 - p_1 < p_2 \leq \sqrt{2N - 3}$ follows from (1). Since $p_1 < p_2$, we have $\frac{2N-p_1}{2N-p_2} = 1 + \frac{p_2 - p_1}{2N - p_2} > 1$. Taking natural logarithms and applying $\ln(1 + x) < x$ for $x > 0$ yields (11).

□

Next, we unify the spectral and graph-theoretic laws governing the Perron eigenvector $\mathbf{u} > \mathbf{0}$ and the maximal prime $p_k = \max(I)$ across all dimensions $k \geq 2$:

Theorem 4.4 (Universal Perron Geometric Mean Ceiling, Maximal Reciprocal Law, and Spectral Cancellation)**.** *Let $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P_\infty^*$ $(k \geq 2)$ be a minimal terminal island governed by the irreducible exponent matrix $M \in \mathbb{Z}_{\geq 0}^{k \times k}$ with spectral radius $\rho(M) \geq 2$ and unique normalized left Perron eigenvector $\mathbf{u} = (u_1, \ldots, u_k)^T > \mathbf{0}$ $(\sum_{i=1}^k u_i = 1,\ \mathbf{u}^T M = \rho(M)\mathbf{u}^T)$. Then:*

1. ***Universal Perron Geometric Mean Ceiling:*** *The $\mathbf{u}$-weighted geometric mean of the primes in I is strictly bounded above by $(2N)^{1/\rho(M)} \leq \sqrt{2N}$:*

$$\sum_{j=1}^{k} u_j \ln p_j = \frac{1}{\rho(M)} \sum_{i=1}^{k} u_i \ln(2N - p_i) < \frac{\ln(2N)}{\rho(M)} \leq \ln \sqrt{2N}, \tag{12}$$

$$\mathcal{G}_{\mathbf{u}}(I) := \prod_{j=1}^{k} p_j^{u_j} < (2N)^{1/\rho(M)} \leq \sqrt{2N}. \tag{13}$$

2. ***Upper-Spectrum Weight Ceiling:*** *Partitioning I at $\sqrt{2N}$ into $J_{Small} = \{j \mid p_j \leq \sqrt{2N}\}$ and $J_{Large} = \{j \mid p_j > \sqrt{2N}\}$, the cumulative Perron weight $U_{Large} = \sum_{j \in J_{Large}} u_j$ satisfies:*

$$U_{Large} < \frac{\frac{2}{\rho(M)} \ln \sqrt{2N} - \ln 3}{\ln \sqrt{2N} - \ln 3} \leq \frac{2}{\rho(M)} \leq 1. \tag{14}$$

3. ***Maximal Prime Reciprocal Identity:*** *By Lemma 3.9 (Part 2), $p_k = \max(I)$ has in-degree 1, dividing a unique complement $2N - p_m = p_k^{a_{m,k}} Q_m$ $(1 \leq m \leq k-1,\ a_{m,k} \geq 1$, $Q_m = \prod_{j=1}^{k-1} p_j^{a_{m,j}} \geq 1$, and $a_{i,k} = 0$ for all $i \neq m)$. Writing $2N - p_k = Q_k = \prod_{j=1}^{k-1} p_j^{a_{k,j}} \geq 9$, the island satisfies the exact Diophantine reciprocal identity:*

$$Q_k - p_m = p_k \left(p_k^{a_{m,k}-1} Q_m - 1 \right) \geq 2p_k, \tag{15}$$

and the left Perron component u_k satisfies the exact decoupling identity:

$$\rho(M) u_k = a_{m,k} u_m \implies u_k = \frac{a_{m,k} u_m}{\rho(M)}. \tag{16}$$

4. ***The Great Perron Cancellation $(a_{m,k} = 1)$:*** *Whenever $a_{m,k} = 1$ (in particular, whenever $p_k > \sqrt{2N-3}$ by Proposition 4.2), we have $Q_m \geq 3$, $\rho(M) u_k = u_m$, and the maximal prime logarithm $\ln p_k$ cancels identically from both sides of the Perron spectral identity $\mathbf{u}^T \mathbf{y} = \rho(M) \mathbf{u}^T \mathbf{x}$, reducing it strictly to the $(k-1)$-prime sub-alphabet $\{p_1, \ldots, p_{k-1}\}$:*

$$\sum_{i \neq m} u_i \ln(2N - p_i) + u_m \ln Q_m = \rho(M) \sum_{j=1}^{k-1} u_j \ln p_j. \tag{17}$$

Proof of Theorem 4.4. 1. **Proof of (1):** From Proposition 3.8, $\mathbf{u}^T \mathbf{x} = \frac{1}{\rho(M)} \mathbf{u}^T \mathbf{y}$. Since $2N - p_i \leq 2N - 3 < 2N$ for all i and $\sum_{i=1}^k u_i = 1$, we have $\mathbf{u}^T \mathbf{y} < \ln(2N)$, yielding (12).

2. **Proof of (2):** Using $\ln p_j \geq \ln 3$ for $j \in J_{\text{Small}}$ and $\ln p_j > \ln \sqrt{2N}$ for $j \in J_{\text{Large}}$, we have:

$$(1 - U_{\text{Large}}) \ln 3 + U_{\text{Large}} \ln \sqrt{2N} \leq \mathbf{u}^T \mathbf{x} < \frac{2}{\rho(M)} \ln \sqrt{2N},$$

which rearranges directly to (14).

3. **Proof of (3):** Equating $2N = p_k + Q_k = p_m + p_k^{a_{m,k}} Q_m$ gives $Q_k - p_m = p_k(p_k^{a_{m,k}-1} Q_m - 1)$. Since $R_m = a_{m,k} + \sum_{j=1}^{k-1} a_{m,j} \geq 2$, the odd product $p_k^{a_{m,k}-1} Q_m$ is at least 3, so $p_k^{a_{m,k}-1} Q_m - 1 \geq 2$. Evaluating the k-th column of $\mathbf{u}^T M = \rho(M)\mathbf{u}^T$ gives $\sum_{i=1}^{k} u_i a_{i,k} = u_m a_{m,k} = \rho(M) u_k$.

4. **Proof of (4):** When $a_{m,k} = 1$, $R_m = \sum_{j=1}^{k} a_{m,j} \geq 2 \implies \sum_{j=1}^{k-1} a_{m,j} \geq 1 \implies Q_m \geq 3$, and (16) gives $\rho(M) u_k = u_m$. Expanding row m on the left of $\sum_{i=1}^{k} u_i \ln(2N - p_i) = \rho(M) \sum_{j=1}^{k} u_j \ln p_j$ gives $u_m \ln(2N - p_m) = u_m \ln p_k + u_m \ln Q_m$, while the $j = k$ term on the right is $\rho(M) u_k \ln p_k = u_m \ln p_k$. Subtracting $u_m \ln p_k$ from both sides yields (17). □

Proposition 4.5 (Universal Sub-Square Semiprime Impossibility)**.** *Let $I = \{p_1 < \cdots < p_k\} \subseteq P_\infty^*$ ($k \geq 2$) be a minimal terminal island. For any row $i \in \{1, \ldots, k\}$, if every prime factor p_j ($j \in S_i$) dividing $2N - p_i$ satisfies $p_j < \sqrt{2N - p_i}$—which holds automatically whenever $p_i \leq \sqrt{2N}$ and $\max_{j \in S_i} p_j \leq \sqrt{2N} - 2$ (and in particular for all $i \in \{1, \ldots, k-1\} \setminus \{m\}$ whenever $p_{k-1} \leq \sqrt{2N} - 2$, as well as for $i = k$ whenever $p_k \leq \sqrt{2N}$ or $p_{k-1} \leq \sqrt{\frac{2}{3}(2N)}$)—then $2N - p_i$ cannot be a semiprime; its row sum must satisfy:*

$$R_i = \sum_{j=1}^{k} a_{i,j} \geq 3. \tag{18}$$

Proof of Proposition 4.5. Suppose for contradiction that $R_i = 2$, so $2N - p_i = p_a p_b$ (allowing $a = b$) with $p_a, p_b \leq \max_{j \in S_i} p_j$. If $\max_{j \in S_i} p_j < \sqrt{2N - p_i}$, then $2N - p_i = p_a p_b < (\sqrt{2N - p_i})^2 = 2N - p_i$, an immediate contradiction. In particular, whenever $p_i \leq \sqrt{2N}$ and $\max_{j \in S_i} p_j \leq \sqrt{2N} - 2$, we have:

$$2N - \sqrt{2N} \leq 2N - p_i = p_a p_b \leq (\sqrt{2N} - 2)^2 = 2N - 4\sqrt{2N} + 4 \implies 3\sqrt{2N} \leq 4,$$

an impossibility for all $2N > 100$. Similarly, for $i = k$, since $p_k \leq \frac{2N-5}{3} < \frac{1}{3}(2N)$, we have $2N - p_k > \frac{2}{3}(2N)$, so $p_{k-1} \leq \sqrt{\frac{2}{3}(2N)}$ implies $\max_{j \in S_k} p_j \leq p_{k-1} < \sqrt{2N - p_k}$, forcing $R_k \geq 3$. □

4.2 Unified Diophantine and Radical Obstructions Across All Dimensions ($k \geq 2$)

By Theorem 4.3, **every** minimal terminal island of dimension $k \geq 2$ generates the coprime proximity equation $(2N - p_1) - (2N - p_2) = p_2 - p_1 < \sqrt{2N}$ with $S_1 \cap S_2 = \emptyset$. We now state the two complementary Diophantine mechanisms—effective linear forms in logarithms (Baker–Matveev–Laurent) and coprime radical bounds (*abc*)—that obstruct this coincidence.

Theorem 4.6 (Baker–Matveev–Laurent Explicit Quantitative Lower Bounds [1, 5, 6])**.** *Let $\alpha_1, \ldots, \alpha_n \in \mathbb{P}$ be distinct prime numbers, and let $b_1, \ldots, b_n \in \mathbb{Z} \setminus \{0\}$ be non-zero integers with $B = \max(3, \max_j |b_j|)$. Then $\Lambda = \sum_{j=1}^{n} b_j \ln \alpha_j \neq 0$ by unique prime factorization, and:*

1. ***Bivariate Forms** ($n = 2$**; Laurent [5]**):*

$$\ln |\Lambda| \geq -17.9 \cdot \left(\max \left\{ \ln \left(\frac{|b_1|}{\ln \alpha_2} + \frac{|b_2|}{\ln \alpha_1} \right) + 0.38,\ 10 \right\} \right)^2 \cdot \ln \alpha_1 \ln \alpha_2. \tag{19}$$

In particular, whenever $\frac{|b_1|}{\ln \alpha_2} + \frac{|b_2|}{\ln \alpha_1} \leq e^{9.62} \approx 15,063$ (which holds for all $B \leq 40$), $\ln |\Lambda| \geq -1790 \ln \alpha_1 \ln \alpha_2$.

2. ***Multivariate Forms*** *($n \geq 3$;* ***Matveev [6]****):*

$$\ln|\Lambda| > -C(n)\ln(eB)\prod_{j=1}^{n}\ln\alpha_j, \quad \textit{where } C(n) = 1.4\cdot 30^{n+3}\cdot n^{4.5}. \tag{20}$$

Proposition 4.7 (Unconditional Elimination in the Bounded-Height Regime ($k \geq 2$))**.** *Let $I = \{p_1 < \cdots < p_k\} \subseteq P_\infty^*$ ($k \geq 2$) be a minimal terminal island, and let $1 \leq i < j \leq k$ be any pair of indices with $p_j \leq \sqrt{2N}$. Let $S_{i,j} = \{r \in \{1,\dots,k\} \mid a_{i,r} \neq a_{j,r}\}$ have cardinality $n = |S_{i,j}| \geq 2$. Then the linear form $\Lambda_{i,j} = \ln\left(\frac{2N-p_i}{2N-p_j}\right) = \sum_{r\in S_{i,j}}(a_{i,r}-a_{j,r})\ln p_r$ satisfies $0 < \Lambda_{i,j} < \frac{p_j-p_i}{2N-p_j} < \frac{2}{\sqrt{2N}}$, and the configuration is* ***unconditionally impossible*** *whenever the prime heights in $S_{i,j}$ satisfy:*

$$\begin{cases} 1790\ln p_a \ln p_b < \ln(2N-p_j) - \ln(p_j-p_i) & \text{if } n = 2 \ (S_{i,j} = \{a,b\},\ B \leq 15000), \\ C(n)\ln(eB)\prod_{r\in S_{i,j}}\ln p_r < \ln(2N-p_j) - \ln(p_j-p_i) & \text{if } n \geq 3. \end{cases} \tag{21}$$

In particular, for any fixed prime subset $\{p_r\}_{r\in S_{i,j}}$, Theorem 4.6 unconditionally eliminates all counterexamples $2N$ above an explicit finite threshold $N_0(\{p_r\})$.

Proof of Proposition 4.7. From $\Lambda_{i,j} = \ln\left(1+\frac{p_j-p_i}{2N-p_j}\right) < \frac{p_j-p_i}{2N-p_j}$, taking natural logarithms gives the upper bound $\ln|\Lambda_{i,j}| < \ln(p_j-p_i) - \ln(2N-p_j)$. Comparing this directly with the lower bounds (19) and (20) of Theorem 4.6 yields an immediate contradiction whenever (21) holds. For any fixed primes $\{p_r\}_{r\in S_{i,j}}$, the left-hand side of (21) grows only as $O(\ln\ln N)$ via $\ln(eB) = O(\ln\ln N)$, whereas the right-hand side grows as $\geq \frac{1}{2}\ln(2N)$, guaranteeing unconditional elimination for all $2N > N_0(\{p_r\})$. □

When the primes in $S_1 \cup S_2$ are not fixed but grow as fractional powers of $2N$ with high exponents $a_{1,j}, a_{2,r} \geq 2$, the coprimality $\gcd(2N-p_1, 2N-p_2) = 1$ of Theorem 4.3 directly activates the radical obstruction of Masser and Oesterlé (*abc*):

Proposition 4.8 (Universal Coprime Radical (*abc*) Obstruction for High-Multiplicity Islands)**.** *Let $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P_\infty^*$ ($k \geq 2$) be a minimal terminal island. By Theorem 4.3, the positive integers:*

$$A = 2N - p_2 = \prod_{r\in S_2} p_r^{a_{2,r}}, \quad B = p_2 - p_1, \quad C = 2N - p_1 = \prod_{j\in S_1} p_j^{a_{1,j}}$$

satisfy $A + B = C$ with $\gcd(A,B,C) = \gcd(2N-p_1, 2N-p_2) = 1$. Their radical satisfies:

$$\operatorname{rad}(ABC) = \operatorname{rad}(p_2-p_1)\prod_{j\in S_1\cup S_2} p_j \leq \frac{p_2-p_1}{2}\prod_{j\in S_1\cup S_2} p_j < \frac{1}{2}p_2\prod_{j\in S_1\cup S_2} p_j < \frac{1}{2}\sqrt{2N}\prod_{j\in S_1\cup S_2} p_j. \tag{22}$$

Consequently, under the Masser–Oesterlé abc conjecture ($C < K_\varepsilon \operatorname{rad}(ABC)^{1+\varepsilon}$), no minimal terminal island of any dimension $k \geq 2$ can exist for sufficiently large $2N$ whenever the base radical product satisfies:

$$p_2\prod_{j\in S_1\cup S_2} p_j \leq (2N)^{1-\delta} \quad \left(\textit{in particular, whenever } \prod_{j\in S_1\cup S_2} p_j \leq (2N)^{\frac{1}{2}-\delta}\right) \quad (\delta > 0).$$

Proof of Proposition 4.8. By Theorem 4.3, $\gcd(A,C) = \gcd(2N-p_2, 2N-p_1) = 1$, which implies $\gcd(A,B) = \gcd(B,C) = 1$. Since $p_1 < p_2$ are odd primes, $B = p_2 - p_1$ is even, so $\operatorname{rad}(B) \leq \frac{p_2-p_1}{2} < \frac{1}{2}p_2 < \frac{1}{2}\sqrt{2N}$. Furthermore, every prime dividing AC has index in $S_1 \cup S_2$, so $\operatorname{rad}(AC) = \prod_{j\in S_1\cup S_2} p_j$, establishing (22). If $p_2\prod_{j\in S_1\cup S_2} p_j \leq (2N)^{1-\delta}$, then $\operatorname{rad}(ABC) < \frac{1}{2}(2N)^{1-\delta}$, which for $\varepsilon < \frac{\delta}{1-\delta}$ violates $2N - \sqrt{2N} < C < K_\varepsilon \operatorname{rad}(ABC)^{1+\varepsilon}$ for all sufficiently large $2N$. □

4.3 Low-Dimensional Specializations: $k = 1$ and $k = 2$

We now specialize our universal invariants to dimensions $k = 1$ and $k = 2$.

Proposition 4.9 (Elimination of Self-Loops: $k = 1$ Collapse)**.** *No prime $p \in P^*(0)$ can belong to its own divisor image $D(\{p\})$. Consequently, no 1-element minimal terminal island can exist ($k = |I| \geq 2$).*

Proof of Proposition 4.9. For any $p \in P^*(0)$, $p \nmid 2N \implies p \nmid (2N - p)$, precluding $p \in D(\{p\})$ and confirming $k = |I| \geq 2$. □

Proposition 4.10 (Structural and Computational Elimination of the $k = 2$ Two-Prime Cycle)**.** *Let $I = \{p_1, p_2\} \subseteq P^*_\infty$ $(3 \leq p_1 < p_2)$ be a hypothetical minimal terminal island of cardinality $k = 2$, governed by:*

$$2N - p_1 = p_2^{a_1} \quad \text{and} \quad 2N - p_2 = p_1^{a_2} \quad (a_1 = a_{1,2} \geq 1,\ a_2 = a_{2,1} \geq 1). \tag{23}$$

Then:

1. ***Linear Exponent Elimination:*** *It is impossible to have $\min(a_1, a_2) = 1$; thus $2 \leq a_1 < a_2$.*

2. ***Exact Multiplier, 2-adic LTE Symmetry, and Master Congruence:*** *There exists a unique even integer $c \geq 2$ such that:*

$$p_2^{a_1-1} - 1 = cp_1, \quad p_1^{a_2-1} - 1 = cp_2, \quad \text{and} \quad 2N = cp_1p_2 + p_1 + p_2, \tag{24}$$

satisfying the exact 2-adic valuation symmetry $v_2(p_2^{a_1-1} - 1) = v_2(c) = v_2(p_1^{a_2-1} - 1)$ and the single-variable polynomial identity:

$$p_1c^{a_1} + c^{a_1-1} = (p_1^{a_2-1} - 1)^{a_1-1} \implies p_1c^{a_1} + c^{a_1-1} \equiv (-1)^{a_1-1} \pmod{p_1^{a_2-1}}. \tag{25}$$

3. ***Unconditional Algebraic Elimination of Small Exponents for All $2N \to \infty$:*** *The exponent pairs $(a_1, a_2) \in \{(2, 3), (2, 4), (2, 5)\}$ possess **no solutions in odd primes $3 \leq p_1 < p_2$ for any $2N \geq 4$.** More generally, when $a_1 = 2$ and $a_2 \geq 6$, $c = mp_1 - 1$ for an odd integer $m \geq 1$ satisfying $p_1^{a_2-2} = (mp_1 - 1)^2 + m$.*

4. ***Complete Elimination Across Empirical, Fixed-Base, and Radical Regimes:***

   - Unconditional Verification up to $2N \leq 10^{24}$ (Covering $4 \cdot 10^{18}$): *For all $2N \leq 10^{24}$, the unique solution to (23) in odd primes is $(p_1, p_2, a_1, a_2) = (3, 13, 3, 7)$ at $2N = 2200$, which is not a Goldbach counterexample (possessing 46 Goldbach partitions).*
   - Unconditional Fixed-Base Elimination (Tijdeman–Laurent): *For any fixed prime pair (p_1, p_2), Proposition 4.7 unconditionally eliminates all exponents:*

$$a_2 > \frac{1790 \ln p_1 \ln p_2 + \ln p_2}{\ln p_1}.$$

   - Radical (abc) Elimination for All $a_1 \geq 3, a_2 \geq 4$: *Since $p_1 < (2N)^{1/a_2}$ and $p_2 < (2N)^{1/a_1}$, the base radical product satisfies $p_2 \prod_{j \in S_1 \cup S_2} p_j = p_1p_2^2 < (2N)^{\frac{2}{a_1} + \frac{1}{a_2}} \leq (2N)^{11/12} = (2N)^{1-1/12}$ for all $a_1 \geq 3, a_2 \geq 4$, so Proposition 4.8 eliminates all $k = 2$ cycles with $a_1 \geq 3, a_2 \geq 4$ under the abc conjecture.*

Proof of Proposition 4.10. 1. **Proof of (1):** If $a_1 = 1$ or $a_2 = 1$, then $2N = p_1 + p_2$, an explicit Goldbach partition contradicting the counterexample hypothesis. Since $3 \leq p_1 < p_2$, we have $p_1^{a_2} = 2N - p_2 < 2N - p_1 = p_2^{a_1}$. Combined with $p_2(p_2^{a_1-1} - 1) = p_1(p_1^{a_2-1} - 1)$ and $p_1 < p_2$, this forces $p_2^{a_1-1} - 1 < p_1^{a_2-1} - 1 < p_2^{a_2-1} - 1 \implies 2 \leq a_1 < a_2$.

2. **Proof of (2):** Subtracting the two equations in (23) gives $p_2(p_2^{a_1-1}-1) = p_1(p_1^{a_2-1}-1)$. Since $\gcd(p_1,p_2)=1$, $c = \frac{p_2^{a_1-1}-1}{p_1} = \frac{p_1^{a_2-1}-1}{p_2}$ is an even positive integer, and $2N = p_1 + p_2(cp_1+1) = cp_1p_2+p_1+p_2$. Because p_1, p_2 are odd, $v_2(p_2^{a_1-1}-1) = v_2(c) = v_2(p_1^{a_2-1}-1)$. Raising $cp_2 = p_1^{a_2-1}-1$ to the power a_1-1 and substituting $p_2^{a_1-1} = cp_1+1$ yields $c^{a_1-1}(cp_1+1) = (p_1^{a_2-1}-1)^{a_1-1}$, proving (25).

3. **Proof of (3):** When $a_1 = 2$, (25) reduces to $p_1c^2 + c = p_1^{a_2-1} - 1 \iff p_1(p_1^{a_2-2} - c^2) = c+1$. Thus $c+1 = mp_1$ for some odd integer $m \geq 1$ ($c = mp_1 - 1$), and substituting into $p_1^{a_2-2} = c^2 + m$ gives:

$$p_1^{a_2-2} = (mp_1-1)^2 + m = m^2p_1^2 - 2mp_1 + (m+1). \tag{26}$$

   - *Case* $(a_1,a_2) = (2,3)$*:* Equation (26) becomes $p_1 = (mp_1-1)^2+m \geq (p_1-1)^2+1 = p_1^2 - 2p_1 + 2 > p_1$ for all $p_1 \geq 3$, a contradiction.
   - *Case* $(a_1,a_2) = (2,4)$*:* Here $a_2 - 2 = 2 \geq 1$, so reducing (26) modulo p_1 forces $p_1 \mid (m+1)$. Since m and p_1 are odd, $m+1 = 2rp_1 \geq 2p_1 \implies m \geq 2p_1 - 1$. Then $p_1^2 = (mp_1-1)^2 + m \geq (2p_1^2 - p_1 - 1)^2 + 2p_1 - 1 > p_1^2$, a contradiction.
   - *Case* $(a_1,a_2) = (2,5)$*:* Substituting $m = 2rp_1 - 1$ ($r \geq 1$) into (26) with $a_2 = 5$ and dividing by p_1 gives $p_1^2 = (2rp_1-1)^2p_1 - 4rp_1 + (2r+2) \implies p_1 \mid 2(r+1) \implies p_1 \mid (r+1) \implies r \geq p_1 - 1 \implies m \geq 2p_1^2 - 2p_1 - 1$. Then $(mp_1-1)^2 + m > p_1^6 > p_1^3$, a contradiction.

4. **Proof of (4):** For $2N \leq 10^{24}$, since $(a_1,a_2) \in \{(2,3),(2,4),(2,5)\}$ are unconditionally eliminated by (3), any solution has either $a_2 \geq 6$ (limiting $p_1 < (10^{24})^{1/6} = 10^4$, i.e., the first $1,229$ odd primes) or $(a_1,a_2) \in \{(3,4),(3,5),(4,5)\}$ (where (25) uniquely determines $c \bmod p_1^{a_2-1}$). Exhaustive inspection confirms that $(p_1,p_2,a_1,a_2) = (3,13,3,7)$ ($2N = 2200$) is the sole prime solution. The fixed-base and radical bounds follow immediately from Propositions 4.7 and 4.8.

□

4.4 Case $k = 3$: Elimination of Three-Prime Cycles

Proposition 4.11 (Collapse of the $k=3$ Three-Prime Cycle)**.** *Let* $I = \{p_1 < p_2 < p_3\} \subset \mathcal{P}_{\leq \frac{2N-5}{3}}$ *be a hypothetical minimal terminal island of cardinality* $k = 3$*, governed by:*

$$2N - p_i = \prod_{j \neq i} p_j^{a_{i,j}}, \quad a_{i,i} = 0, \quad R_i = \sum_{j=1}^{3} a_{i,j} \geq 2 \quad (\forall i \in \{1,2,3\}). \tag{27}$$

Then:

1. ***Unconditional Elimination of Square-Free Cycles*** *($a_{i,j} \in \{0,1\}$):* *No square-free* $k=3$ *island can exist for any* $2N > 100$.

2. ***Canonical In-Degree Reduction:*** *By Lemma 3.9(2), the maximal prime* $p_3 = \max(I)$ *has* $\text{in-deg}(p_3) = 1$*, dividing* $2N - p_m$ *for a unique parent* $m \in \{1,2\}$ *and forcing the other base complement to be a pure prime power:*

   - Branch A ($m = 2$, $p_3 \mid (2N - p_2)$):

$$2N - p_1 = p_2^{a_{1,2}} \quad (a_{1,2} \geq 2), \quad 2N - p_2 = p_1^{a_{2,1}} p_3^{a_{2,3}} \quad (a_{2,3} \geq 1), \quad 2N - p_3 = p_1^{a_{3,1}} p_2^{a_{3,2}}. \tag{28}$$

- Branch B ($m = 1$, $p_3 \mid (2N - p_1)$):

$$2N - p_1 = p_2^{a_{1,2}} p_3^{a_{1,3}} \quad (a_{1,3} \geq 1), \quad 2N - p_2 = p_1^{a_{2,1}} \quad (a_{2,1} \geq 2), \quad 2N - p_3 = p_1^{a_{3,1}} p_2^{a_{3,2}}. \tag{29}$$

3. ***Exact Multiplier Identity for Branched Cycles:*** *In Branch A ($m = 2$), whenever $a_{3,2} \geq 1$, we have $p_3 - p_1 = 2rp_2^{a_{3,2}}$ ($r \geq 1$), and there exists an even integer $K \geq 2$ such that $p_1^{a_{2,1}+a_{2,3}-1} - 1 = Kp_2$ ($a_{2,1} + a_{2,3} \geq 3$). When $a_{2,3} = 1$, K simultaneously satisfies the exact dual identity:*

$$p_1^{a_{2,1}} - 1 = Kp_2, \quad p_1^{a_{3,1}} p_2^{a_{3,2}-1} - 1 = Kp_3, \quad \text{and} \quad Kp_2^{a_{1,2}} + Kp_1 + 1 = p_1^{a_{2,1}+a_{3,1}} p_2^{a_{3,2}-1}, \tag{30}$$

which unconditionally eliminates $a_{2,1} = 1$ and enforces the strict congruence:

$$K^{a_{1,2}-1}(Kp_1 + 1) \equiv (-1)^{a_{1,2}-1} \pmod{p_1^{a_{2,1}+a_{3,1}}}.$$

(Symmetrically, in Branch B with $a_{3,1} \geq 1$ and $a_{1,3} = 1$, there exists an even integer $K' \geq 2$ satisfying $p_2^{a_{1,2}} - 1 = K'p_1$, $p_1^{a_{3,1}-1} p_2^{a_{3,2}} - 1 = K'p_3$, and $K'^{a_{2,1}-1}(K'p_2 + 1) \equiv (-1)^{a_{2,1}-1} \pmod{p_2^{a_{1,2}+a_{3,2}}}$.)

4. ***Elimination Across Bounded-Height and Root-Compressed Radical (abc) Regimes:*** *Every $k = 3$ cycle satisfying (21) is unconditionally eliminated in the bounded-height regime by Proposition 4.7, and every root-compressed high-multiplicity configuration satisfying $p_2 \prod_{j \in S_1 \cup S_2} p_j \leq (2N)^{1-\delta}$ (such as pure 3-cycles with $\frac{2}{a_{1,2}} + \frac{1}{a_{2,3}} < 1$ or branched cycles with $\frac{2}{a_{1,2}} + \frac{1}{\min(a_{2,1}, a_{2,3})} < 1$) is eliminated conditionally under the abc conjecture by Proposition 4.8.*

Proof of Proposition 4.11. 1. **Proof of (1) (Square-Free Products $a_{i,j} \in \{0, 1\}$):** Since $a_{i,i} = 0$ and $R_i = \sum_{j=1}^{3} a_{i,j} \geq 2$, every off-diagonal entry must equal 1:

$$2N - p_1 = p_2 p_3, \quad 2N - p_2 = p_1 p_3, \quad 2N - p_3 = p_1 p_2. \tag{31}$$

Subtracting $2N - p_2$ from $2N - p_1$ gives $p_2 - p_1 = p_3(p_2 - p_1)$. Dividing by $p_2 - p_1 > 0$ forces $p_3 = 1$, contradicting $p_3 > p_2 > p_1 \geq 3 \implies p_3 \geq 7$.

2. **Proof of (2) (Maximal Prime In-Degree Reduction):** By Lemma 3.9 (Part 2), $p_3 = \max(I)$ has in-degree 1 and divides at most one of $\{2N - p_1, 2N - p_2\}$. If $m = 2$ ($p_3 \mid (2N - p_2)$ and $p_3 \nmid (2N - p_1)$), the only available prime factor for $2N - p_1$ in $I \setminus \{p_1\}$ is p_2, forcing $2N - p_1 = p_2^{a_{1,2}}$ with $a_{1,2} \geq 2$ and establishing (28). Subtracting $2N - p_3$ from $2N - p_1$ in Branch A yields:

$$p_3 - p_1 = p_2^{a_{1,2}} - p_1^{a_{3,1}} p_2^{a_{3,2}} = p_2^{a_{3,2}} \left(p_2^{a_{1,2}-a_{3,2}} - p_1^{a_{3,1}} \right) > 0 \implies a_{1,2} \geq a_{3,2}. \tag{32}$$

If $m = 1$ ($p_3 \mid (2N - p_1)$ and $p_3 \nmid (2N - p_2)$), the only available prime factor for $2N - p_2$ in $I \setminus \{p_2\}$ is p_1, forcing $2N - p_2 = p_1^{a_{2,1}}$ with $a_{2,1} \geq 2$ and establishing (29).

3. **Proof of (3) (Branched Cycles):** In Branch A ($m = 2$) with $a_{3,2} \geq 1$, (32) implies $p_2^{a_{3,2}} \mid (p_3 - p_1)$. Since p_1, p_3 are odd primes and p_2 is odd, $p_3 - p_1 = 2rp_2^{a_{3,2}}$ ($r \geq 1$), so $p_3 \equiv p_1 \pmod{p_2}$. Subtracting $2N - p_2$ from $2N - p_1$ modulo p_2 gives:

$$p_2 - p_1 = p_2^{a_{1,2}} - p_1^{a_{2,1}} p_3^{a_{2,3}} \implies p_1^{a_{2,1}+a_{2,3}} \equiv p_1^{a_{2,1}} p_3^{a_{2,3}} \equiv p_1 \pmod{p_2}.$$

Since $\gcd(p_1, p_2) = 1$, we have $p_1^{a_{2,1}+a_{2,3}-1} - 1 = Kp_2$ for an even integer $K \geq 2$, which forces $a_{2,1} + a_{2,3} \geq 3$ because $p_1 < p_2$.

When $a_{2,3} = 1$, this gives $p_1^{a_{2,1}} - 1 = Kp_2$ ($a_{2,1} \geq 2$, since $a_{2,1} = 1 \implies p_2 = \frac{p_1-1}{K} < p_1$, contradicting $p_1 < p_2$). By the Maximal Prime Reciprocal Identity (Theorem 4.4(3), (15)) with $Q_3 = p_1^{a_{3,1}} p_2^{a_{3,2}}$ and $Q_2 = p_1^{a_{2,1}}$:

$$p_2\big(p_1^{a_{3,1}} p_2^{a_{3,2}-1} - 1\big) = Q_3 - p_2 = p_3(Q_2 - 1) = p_3(p_1^{a_{2,1}} - 1) = Kp_2p_3.$$

Dividing by p_2 yields $p_1^{a_{3,1}} p_2^{a_{3,2}-1} - 1 = Kp_3$. Multiplying $p_2^{a_{1,2}} + p_1 - p_2 = p_1^{a_{2,1}} p_3$ by K and substituting $Kp_2 = p_1^{a_{2,1}} - 1$ and $Kp_3 = p_1^{a_{3,1}} p_2^{a_{3,2}-1} - 1$ cancels $-p_1^{a_{2,1}}$ on both sides, establishing (30). The symmetric relations in Branch B follow by interchanging indices 1 and 2.

4. **Proof of (4) (Bounded-Height and Radical Elimination):** By Theorem 4.3, $A = 2N - p_2$, $B = p_2 - p_1$, and $C = 2N - p_1$ form a coprime triple $A + B = C$ with $0 < \Lambda_{1,2} < \frac{2}{\sqrt{2N}}$.

   - For fixed or bounded primes $\{p_1, p_2, p_3\}$ satisfying (21), Proposition 4.7 unconditionally eliminates the cycle.
   - In Branch A, $p_2 = (2N - p_1)^{1/a_{1,2}} < (2N)^{1/a_{1,2}}$. For the pure 3-cycle ($a_{2,1} = a_{3,2} = 0$), $\prod_{j \in S_1 \cup S_2} p_j = p_2p_3 < (2N)^{\frac{1}{a_{1,2}} + \frac{1}{a_{2,3}}}$, so $p_2 \prod_{j \in S_1 \cup S_2} p_j < (2N)^{\frac{2}{a_{1,2}} + \frac{1}{a_{2,3}}}$, which is eliminated by Proposition 4.8 whenever $\frac{2}{a_{1,2}} + \frac{1}{a_{2,3}} < 1$. When $a_{2,1} \geq 1$, $(p_1p_3)^{\min(a_{2,1}, a_{2,3})} \leq p_1^{a_{2,1}} p_3^{a_{2,3}} < 2N$, so $p_2(p_1p_2p_3) < (2N)^{\frac{2}{a_{1,2}} + \frac{1}{\min(a_{2,1}, a_{2,3})}}$, which is eliminated by Proposition 4.8 whenever $\frac{2}{a_{1,2}} + \frac{1}{\min(a_{2,1}, a_{2,3})} < 1$ (and symmetrically in Branch B).

□

4.5 Square-Free Elimination and Perron Component Bounds ($k \geq 4$)

Proposition 4.12 (Binary Exponent Domain Collapse ($a_{i,j} \in \{0, 1\}$))**.** *For any $k \geq 4$, no minimal terminal island $I = \{p_1 < p_2 < \cdots < p_k\} \subset \mathcal{P}_{\leq \frac{2N-5}{3}}$ can exist with all exponents restricted to $a_{i,j} \in \{0, 1\}$ (square-free products) in the following fundamental regimes:*

1. ***Uniformly Dense Square-Free Islands*** *($|S_i| = k - 1$* ***for all*** *$i \in \{1, \ldots, k\}$)**:** *Unconditionally impossible for all $k \geq 4$.*

2. ***Four-Prime Square-Free Islands*** *($k = 4$)**:** Unconditionally impossible across all dense and sparse graph topologies.*

3. ***Symmetric Cofactor Subsystems*** *($k \geq 5$)**:** Any square-free island containing two nodes $p_a < p_b$ sharing an identical cofactor $Q = \prod_{r \notin \{a,b\}} p_r^{a_{a,r}} = \prod_{r \notin \{a,b\}} p_r^{a_{b,r}}$ ($2N - p_a = p_bQ$ and $2N - p_b = p_aQ$) is unconditionally impossible.*

Proof of Proposition 4.12. Let $I = \{p_1 < p_2 < \cdots < p_k\} \subset \mathcal{P}_{\leq \frac{2N-5}{3}}$ be governed by a binary exponent matrix $a_{i,j} \in \{0, 1\}$, so that $2N - p_i = \prod_{j \in S_i} p_j$ with $S_i \subseteq \{1, \ldots, k\} \setminus \{i\}$ and $R_i = |S_i| \geq 2$.

1. **Case 1 (Uniformly Dense Graph $|S_i| = k - 1$ and Symmetric Cofactors):** If $|S_i| = k - 1$ for all i, then $2N - p_1 = p_2 \prod_{r=3}^{k} p_r$ and $2N - p_2 = p_1 \prod_{r=3}^{k} p_r$. More generally, if any two nodes $p_a < p_b$ satisfy $2N - p_a = p_bQ$ and $2N - p_b = p_aQ$ with $Q \geq 3$, subtracting yields:

$$p_b - p_a = (2N - p_a) - (2N - p_b) = Q(p_b - p_a) \implies Q = 1, \tag{33}$$

contradicting $Q \geq 3$ (and for the dense graph, $Q = \prod_{r=3}^{k} p_r \geq 7 \cdot 11 = 77$).

2. **Case 2 (Complete Polynomial Elimination of Sparse Graphs for $k = 4$):** For $k = 4$, $|S_i| \in \{2, 3\}$ for all $i \in \{1, 2, 3, 4\}$. By Lemma 3.9 (Part 2), $p_4 = \max(I)$ has in-degree 1, so $4 \in S_m$ for a unique $m \in \{1, 2, 3\}$. We branch on $|S_4| \in \{2, 3\}$:

   - *Subcase 2.a ($|S_4| = 2$):* Let $S_4 = \{a, b\} \subset \{1, 2, 3\}$, so $2N - p_4 = p_a p_b$.
     (a) If $|S_m| = 2$ and $m \in \{a, b\}$ (say $m = a$), then $2N - p_a = p_4 p_j$ ($j \neq a$). Equating $2N = p_4 + p_a p_b = p_a + p_4 p_j$ gives:
     $$p_4(p_j - 1) = p_a(p_b - 1) \implies p_4 \mid (p_b - 1) \implies p_b \geq p_4 + 1 > p_4, \tag{34}$$
     contradicting $p_4 = \max(I) > p_b$.
     (b) If $|S_m| = 2$ and $m \notin \{a, b\}$, then $\{a, b, m\} = \{1, 2, 3\}$ and $2N - p_m = p_4 p_a$ (up to relabeling $a \leftrightarrow b$). Equating $2N = p_4 + p_a p_b = p_m + p_4 p_a$ gives:
     $$p_4 - p_m = p_a(p_4 - p_b). \tag{35}$$
     Since $4 \notin S_b$ (as in-deg$(p_4) = 1$) and $|S_b| \geq 2$ with $S_b \subseteq \{a, m\}$, we must have $S_b = \{a, m\}$, so $2N - p_b = p_a p_m$. Subtracting $2N - p_b = p_a p_m$ from $2N - p_4 = p_a p_b$ yields:
     $$p_4 - p_b = p_a(p_m - p_b). \tag{36}$$
     Substituting (36) into (35) gives $p_4 - p_m = p_a^2(p_m - p_b)$. Subtracting (36) from this equation yields:
     $$-(p_m - p_b) = (p_a^2 - p_a)(p_m - p_b) \iff (p_a^2 - p_a + 1)(p_m - p_b) = 0. \tag{37}$$
     Since $p_a \geq 3 \implies p_a^2 - p_a + 1 \geq 7 > 0$, this forces $p_m = p_b$, contradicting distinctness.
     (c) If $|S_m| = 3$, then $S_m = \{1, 2, 3, 4\} \setminus \{m\}$. If $m \notin \{a, b\}$, then $S_m = \{a, b, 4\}$ and $2N - p_m = p_4 p_a p_b$; equating $2N = p_4 + p_a p_b = p_m + p_4 p_a p_b$ gives:
     $$p_4 - p_m = p_a p_b(p_4 - 1) \geq 15(p_4 - 1) > p_4 \implies p_m < 0, \tag{38}$$
     an impossibility. If instead $m \in \{a, b\}$, let c be the unique index in $\{1, 2, 3\} \setminus \{a, b\}$. Since in-deg$(p_4) = 1$, $4 \notin S_c$, and since $c \notin S_c$ and $|S_c| \geq 2$, we are forced to have $S_c = \{a, b\}$, yielding $2N - p_c = p_a p_b = 2N - p_4 \implies p_c = p_4$, contradicting distinctness.
   - *Subcase 2.b ($|S_4| = 3$):* Here $2N - p_4 = p_1 p_2 p_3$. Since in-deg$(p_4) = 1$ ($4 \in S_m$ only for m), for the two nodes $r \in \{1, 2, 3\} \setminus \{m\}$ we have $4 \notin S_r$, and since $|S_r| \geq 2$ and $S_r \subseteq \{1, 2, 3\} \setminus \{r\}$, both nodes $r \neq m$ must have $|S_r| = 2$ with $S_r = \{1, 2, 3\} \setminus \{r\}$. Letting $\{a, b\} = \{1, 2, 3\} \setminus \{m\}$, we have $2N - p_a = p_b p_m$ and $2N - p_b = p_a p_m$. Subtracting gives $p_b - p_a = p_m(p_b - p_a) \implies p_m = 1$, contradicting $p_m \geq 3$.

□

Lemma 4.13 (Perron Component Floor)**.** *In an irreducible non-negative integer matrix $M \in \mathbb{Z}_{\geq 0}^{k \times k}$ with row sums $R_i = \sum_{j=1}^{k} a_{i,j} \geq 2$, every component u_j of the normalized left Perron eigenvector $\mathbf{u} > \mathbf{0}$ ($\sum_{i=1}^{k} u_i = 1$) satisfies:*

$$u_j \geq \frac{1}{k \cdot \rho(M)^{k-1}} \equiv \delta(k, \rho) > 0 \quad (\forall j \in \{1, \ldots, k\}). \tag{39}$$

Proof. Since $\sum_{i=1}^{k} u_i = 1$, the maximal component satisfies $\max_{1 \leq i \leq k} u_i \geq \frac{1}{k}$. By irreducibility, there exists a directed path of length $\ell \leq k - 1$ from the maximal-weight node i to any node j with $(M^\ell)_{i,j} \geq 1$. The eigenvector relation $\mathbf{u}^T M^\ell = \rho(M)^\ell \mathbf{u}^T$ enforces:

$$u_j = \frac{1}{\rho(M)^\ell} \sum_{r=1}^{k} u_r (M^\ell)_{r,j} \geq \frac{u_i (M^\ell)_{i,j}}{\rho(M)^\ell} \geq \frac{1}{k \cdot \rho(M)^{k-1}} = \delta(k, \rho) > 0.$$

□

4.6 Cases $k = 4$ and $k = 5$: Descending In-Degree Cascade and Structural Collapse

We now apply the Descending In-Degree Rigidity of Lemma 3.9 together with our unified Theorems 4.3 and 4.4 to dimensions $k = 4$ and $k = 5$.

Proposition 4.14 (Structural Collapse of $k = 4$ Fixed-Point Islands)**.** *Let $I = \{p_1 < p_2 < p_3 < p_4\} \subset \mathcal{P}_{\leq \frac{2N-5}{3}}$ be a minimal terminal island of cardinality $k = 4$ governed by the irreducible exponent matrix $M \in \mathbb{Z}_{\geq 0}^{4\times 4}$. Then:*

1. ***Non-Binary Multiplicity Floor:*** *By Proposition 4.12(2), M cannot be binary; at least one entry satisfies $a_{i,j} \geq 2$.*

2. ***Base Pure-Power Reduction:*** *By Lemma 3.9(2)–(3), $\text{in-deg}(p_4) = 1$ (entering a unique node p_m, $m \in \{1, 2, 3\}$) and p_3 divides at most one of $\{2N - p_1, 2N - p_2\}$. Consequently:*

   - *If $m = 3$ ($p_4 \mid (2N - p_3)$), then p_4 divides neither $2N - p_1$ nor $2N - p_2$, and p_3 divides at most one of them, forcing at least one base complement to be a **pure prime power**:*

   $$2N - p_1 = p_2^{a_{1,2}} \quad (a_{1,2} \geq 2) \quad \text{or} \quad 2N - p_2 = p_1^{a_{2,1}} \quad (a_{2,1} \geq 2). \tag{40}$$

   - *If $m \in \{1, 2\}$ (p_4 divides $2N - p_1$ or $2N - p_2$), then $p_4 \nmid (2N - p_3)$, so $2N - p_3 = p_1^{a_{3,1}} p_2^{a_{3,2}}$ factors entirely over $\{p_1, p_2\}$, while the remaining base node $r \in \{1, 2\} \setminus \{m\}$ has support $S_r \subseteq \{3 - r, 3\}$.*

3. ***Maximal Reciprocal & Coprime Base Reduction:*** *At node p_4, $2N - p_4 = Q_4 = \prod_{j=1}^{3} p_j^{a_{4,j}}$ satisfies $Q_4 - p_m = p_4(p_4^{a_{m,4}-1} Q_m - 1) \implies Q_4 \equiv p_m \pmod{p_4}$, and the coprime base equation $\prod_{j\in S_1} p_j^{a_{1,j}} - \prod_{r\in S_2} p_r^{a_{2,r}} = p_2 - p_1 < \sqrt{2N}$ has disjoint support of total size $|S_1 \cup S_2| \leq 4$ (and $|S_1 \cup S_2| \leq 3$ whenever $m = 3$), which is eliminated unconditionally in the bounded-height regime of Proposition 4.7 and conditionally under the abc conjecture in the root-compressed high-multiplicity regime $p_2 \prod_{j\in S_1\cup S_2} p_j \leq (2N)^{1-\delta}$ of Proposition 4.8.*

Proof of Proposition 4.14. Assertion (1) is proved in Proposition 4.12(2). For Assertion (2), Lemma 3.9(2) guarantees a unique $m \in \{1, 2, 3\}$ with $a_{m,4} \geq 1$ and $a_{i,4} = 0$ for all $i \neq m$, while Lemma 3.9(3) guarantees $a_{1,3}a_{2,3} = 0$. When $m = 3$, $S_1 \subseteq \{2, 3\}$ and $S_2 \subseteq \{1, 3\}$ with $3 \notin S_1 \cap S_2$, so either $S_1 = \{2\}$ ($2N - p_1 = p_2^{a_{1,2}}$) or $S_2 = \{1\}$ ($2N - p_2 = p_1^{a_{2,1}}$). When $m \in \{1, 2\}$, $a_{3,4} = 0$ and $a_{3,3} = 0$ force $S_3 \subseteq \{1, 2\}$, while for $r \in \{1, 2\} \setminus \{m\}$, $a_{r,4} = a_{r,r} = 0 \implies S_r \subseteq \{m, 3\}$.

For Assertion (3), Theorem 4.4(3) gives $Q_4 - p_m = p_4(p_4^{a_{m,4}-1} Q_m - 1) \geq 2p_4$, whence $\prod_{j=1}^{3} p_j^{a_{4,j}} \geq 2p_4 + p_m$. Furthermore, because $\text{in-deg}(p_4) = 1$ and $a_{1,3}a_{2,3} = 0$, each of p_3, p_4 can appear in at most one of S_1, S_2. Thus the coprime base identity $(2N-p_1)-(2N-p_2) = p_2-p_1$ of Theorem 4.3 involves at most $|S_1\cup S_2| \leq 4$ primes (and at most 3 primes whenever $m = 3$), which is eliminated unconditionally in the bounded-height regime by Proposition 4.7 and conditionally under the *abc* conjecture when $p_2 \prod_{j\in S_1\cup S_2} p_j \leq (2N)^{1-\delta}$ by Proposition 4.8. □

Proposition 4.15 (Structural Collapse of $k = 5$ Fixed-Point Islands)**.** *Let $I = \{p_1 < p_2 < p_3 < p_4 < p_5\} \subset \mathcal{P}_{\leq \frac{2N-5}{3}}$ be a minimal terminal island of cardinality $k = 5$ governed by the irreducible exponent matrix $M \in \mathbb{Z}_{\geq 0}^{5\times 5}$. Then:*

1. ***Descending In-Degree Cascade on $\{p_3, p_4, p_5\}$:*** *By Lemma 3.9(2)–(3):*

   - *$p_5 = \max(I)$ divides a unique complement $2N - p_m$ ($m \in \{1, 2, 3, 4\}$);*

- *p_4 divides at most one complement among $\{2N-p_1, 2N-p_2, 2N-p_3\}$;*
- *p_3 divides at most one complement between $\{2N-p_1, 2N-p_2\}$.*

Consequently, the upper spectrum $\{p_3, p_4, p_5\}$ supplies at most 3 incoming edges in total to the two base complements $\{2N-p_1, 2N-p_2\}$ (and at most 2 incoming edges whenever $m \in \{3,4\}$).

2. ***Base Support Compression:*** *By the Pigeonhole Principle, at least one of the two coprime base complements $2N-p_1$ or $2N-p_2$ has prime support of cardinality $|S_r| \le 2$ (and is forced into a pure prime power $2N-p_1 = p_2^{a_{1,2}}$ or $2N-p_2 = p_1^{a_{2,1}}$ whenever $\{p_3, p_4, p_5\}$ all enter the other base node or $\{p_3, p_4\}$).*

3. ***Reciprocal Floor and Coprime Base Reduction:*** *At node p_5, $Q_5 = 2N - p_5 = \prod_{j=1}^{4} p_j^{a_{5,j}}$ satisfies $Q_5 - p_m = p_5(p_5^{a_{m,5}-1}Q_m - 1) \ge 2p_5$, while the coprime base proximity equation $\prod_{j\in S_1} p_j^{a_{1,j}} - \prod_{r\in S_2} p_r^{a_{2,r}} = p_2 - p_1 < \sqrt{2N}$ of Theorem 4.3 is eliminated unconditionally in the bounded-height regime of Proposition 4.7 and conditionally under the abc conjecture in the root-compressed high-multiplicity regime $p_2 \prod_{j\in S_1\cup S_2} p_j \le (2N)^{1-\delta}$ of Proposition 4.8.*

Proof of Proposition 4.15. By Lemma 3.9, each of p_3, p_4, p_5 can divide at most one of $\{2N-p_1, 2N-p_2\}$, so $|(S_1\cup S_2)\cap\{3,4,5\}| \le 3$. Since $S_1\cap S_2 = \emptyset$ (Theorem 4.3), we have $|S_1\cap\{3,4,5\}| + |S_2\cap\{3,4,5\}| \le 3$, forcing $\min\big(|S_1\cap\{3,4,5\}|, |S_2\cap\{3,4,5\}|\big) \le 1$. Because $S_1 \subseteq \{2,3,4,5\}$ and $S_2 \subseteq \{1,3,4,5\}$, at least one of S_1, S_2 has total cardinality ≤ 2, and if $|S_r\cap\{3,4,5\}| = 0$, that base complement is a pure prime power $p_2^{a_{1,2}}$ or $p_1^{a_{2,1}}$. Combining this with the reciprocal identity (15) of Theorem 4.4(3) and Propositions 4.7–4.8 completes the proof. □

4.7 Case $k \ge 6$: Inductive Node Contraction and Sub-Square Spectral Elimination

Having established the universal coprime base squeeze ($p_2 < \sqrt{2N}$ and $0 < \Lambda_{1,2} < \frac{2}{\sqrt{2N}}$) for all $k \ge 2$ in Theorem 4.3, we now analyze the global exponent mass and sub-square spectral structure for $k \ge 6$.

Proposition 4.16 (Inductive Schur Node Contraction and Exponent Mass Accumulation)**.** *Let $I = \{p_1 < p_2 < \cdots < p_k\} \subset \mathcal{P}_{\le \frac{2N-5}{3}}$ ($k \ge 6$) be a minimal terminal island governed by the irreducible exponent matrix $M \in \mathbb{Z}_{\ge 0}^{k\times k}$ with row sums $R_i = \sum_{j=1}^{k} a_{i,j} \ge 2$. Assume $a_{m,k} = 1$ at the unique incoming parent node p_m of $p_k = \max(I)$ (which holds unconditionally whenever $p_k > \sqrt{2N-3}$ by Proposition 4.2). Then:*

1. ***Single-Step Schur Contraction*** *($k \to k-1$)**:** Algebraic elimination of $p_k = \frac{2N-p_m}{Q_m}$ (where $Q_m = \prod_{j=1}^{k-1} p_j^{a_{m,j}} \ge 3$) from $2N - p_k = \prod_{j=1}^{k-1} p_j^{a_{k,j}}$ yields the exact contracted identity over $\{p_1, \ldots, p_{k-1}\}$:*

$$2N(Q_m - 1) + p_m = Q_m \prod_{j=1}^{k-1} p_j^{a_{k,j}} = \prod_{j=1}^{k-1} p_j^{a_{m,j}+a_{k,j}}. \tag{41}$$

Defining the contracted $(k-1)\times(k-1)$ exponent matrix $M' = (a'_{i,j})$ on $\{p_1, \ldots, p_{k-1}\}$ by $a'_{m,j} = a_{m,j} + a_{k,j}$ and $a'_{i,j} = a_{i,j}$ for $i \in \{1, \ldots, k-1\} \setminus \{m\}$, row m's sum increases to $R'_m = R_m + R_k - 1 \ge R_m + 1 \ge 3$, and the grand exponent sum of M' satisfies:

$$\sum_{i=1}^{k-1}\sum_{j=1}^{k-1} a'_{i,j} = \sum_{i=1}^{k}\sum_{j=1}^{k} a_{i,j} - 1 \ge 2k - 1. \tag{42}$$

2. ***Inductive Contraction to Dimension*** $d = 5$ ***($k \to 5$):*** *If each successive maximal prime p_ℓ ($\ell = k, k-1, \ldots, 6$) has a single incoming parent with unit exponent in the contracted system, iterating (41) $k-5$ times compresses the exponent mass into a 5×5 matrix $M^{(5)} = (A_{i,j})$ on $\{p_1, \ldots, p_5\}$ with row sums $R_i^{(5)} := \sum_{j=1}^{5} A_{i,j}$ satisfying:*

$$\sum_{i=1}^{5} R_i^{(5)} = \sum_{i=1}^{k} R_i - (k-5) \geq k+5, \tag{43}$$

and in the non-semiprime regime where at least $k-1$ rows of M satisfy $R_i \geq 3$ ($\sum_{i=1}^{k} R_i \geq 3k-1$),

$$\sum_{i=1}^{5} R_i^{(5)} \geq (3k-1) - (k-5) = 2k+4 > 2k \implies \overline{R}_5 := \frac{1}{5}\sum_{i=1}^{5} R_i^{(5)} \geq \frac{2k+4}{5} > \frac{2k}{5}, \tag{44}$$

forcing at least one row $r \in \{1, \ldots, 5\}$ to carry an accumulated exponent sum $R_r^{(5)} \geq \lceil (2k+4)/5 \rceil$.

3. ***Uncontracted Base Ceiling and Conditional Dimension Cutoff:*** *For every row $i \in \{1, \ldots, 5\}$ that does not serve as a parent node during contraction, its product remains strictly equal to $2N - p_i < 2N$, bounding its row sum by $R_i^{(5)} = R_i \leq \lfloor \frac{\ln(2N-3)}{\ln 3} \rfloor$ (whereas for any contracted parent row m, the left-hand side increases to $2N(Q_m - 1) + p_m$). In the non-semiprime regime (44), if the contracted average row sum satisfies $\overline{R}_5 \leq \lfloor \frac{\ln(2N-3)}{\ln 3} \rfloor$, then:*

$$k < \frac{5}{2}\left\lfloor \frac{\ln(2N-3)}{\ln 3} \right\rfloor \quad \big(\text{giving } k \leq 97 \text{ at } 2N \approx 4 \cdot 10^{18}\big). \tag{45}$$

Proof of Proposition 4.16. By Lemma 3.9(2), p_k divides only $2N - p_m$. With $a_{m,k} = 1$, we have $2N - p_m = p_k Q_m$ where $Q_m = \prod_{j=1}^{k-1} p_j^{a_{m,j}} \geq 3$ (since $R_m = 1 + \sum_{j=1}^{k-1} a_{m,j} \geq 2$). Substituting $p_k = \frac{2N-p_m}{Q_m}$ into $2N - p_k = \prod_{j=1}^{k-1} p_j^{a_{k,j}}$ and multiplying by Q_m yields (41). Since $a_{k,k} = 0$ and $a_{i,k} = 0$ for all $i \neq m$, removing row k (with sum $R_k = \sum_{j=1}^{k-1} a_{k,j} \geq 2$) and column k (with single non-zero entry $a_{m,k} = 1$) while adding $a_{k,j}$ to row m gives $R'_m = (R_m - 1) + R_k \geq R_m + 1 \geq 3$ and $\sum_{i=1}^{k-1} R'_i = \sum_{i=1}^{k} R_i - 1 \geq 2k-1$, proving (42). Iterating this single-parent elimination $k-5$ times subtracts 1 from the total exponent sum at each step, yielding (43)–(45). □

Remark 4.17 (The Asymptotic Dimension Boundary and the Spectral Decoupling Frontier)**.** In low support dimensions $n = |S_1 \cup S_2|$, Theorem 4.6 has a fixed dimensional constant $C(n)$, whereas as $k \to \infty$ Matveev's constant scales as $C(k) = 1.4 \cdot 30^{k+3} k^{4.5}$. Theorem 4.3 and Proposition 4.16 show that any hypothetical Goldbach counterexample $2N > 4 \cdot 10^{18}$ is forced into an ultra-rigid bottleneck: it must generate a strongly connected digraph with $3 \leq p_1 < p_2 < \sqrt{2N}$, $\gcd(2N - p_1, 2N - p_2) = 1$, and $0 < \Lambda_{1,2} < \frac{2}{\sqrt{2N}} < 10^{-9}$.

Proposition 4.18 (Sub-Square Semiprime Elimination and Spectral Radius Elevation)**.** *Let $2N > 4 \cdot 10^{18}$ be a hypothetical Goldbach counterexample, and let $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P_\infty^*$ ($k \geq 2$) be a minimal terminal island governed by $M \in \mathbb{Z}_{\geq 0}^{k \times k}$ with row sums $R_i = \sum_{j=1}^{k} a_{i,j}$ and left Perron eigenvector $\mathbf{u} > \mathbf{0}$. Let p_m be the unique incoming parent of $p_k = \max(I)$. Then:*

1. ***Sub-Square Semiprime Impossibility*** ***($p_{k-1} \leq \sqrt{2N} - 2$):*** *Whenever $p_{k-1} \leq \sqrt{2N} - 2$, every row $i \in \{1, \ldots, k\} \setminus \{m\}$ satisfies $R_i \geq 3$. Furthermore, if $p_k \leq \sqrt{2N}$ and $a_{m,k} = 1$ (or unconditionally if $p_k \leq \sqrt{2N} - 2$), then* ***every row without exception*** *satisfies $R_i \geq 3$ ($\forall i \in \{1, \ldots, k\}$).*

2. ***Spectral Radius Elevation and Perron Geometric Mean Compression*** *($p_k \le \sqrt{2N}$, $a_{m,k} = 1$): Whenever $R_i \ge 3$ for all $i \in \{1, \dots, k\}$, the Collatz–Wielandt theorem enforces $\rho(M) \ge \min_i R_i \ge 3$, compressing the Perron weighted geometric mean ceiling (Theorem 4.4(1)) to:*

$$\mathcal{G}_{\mathbf{u}}(I) = \prod_{j=1}^{k} p_j^{u_j} < (2N)^{1/\rho(M)} \le (2N)^{1/3} \quad \left(\approx 1.58 \times 10^6 \text{ at } 2N \approx 4 \cdot 10^{18}\right), \tag{46}$$

which forces $p_1 < (2N)^{1/3}$ and eliminates any island with $\mathcal{G}_{\mathbf{u}}(I) \ge (2N)^{1/3}$. More generally, if all primes in I are bounded by $p_k \le (2N)^{1/s}$ for an integer $s \ge 2$, then no complement $2N - p_i$ can have $R_i \le s$ unless $i = m$ with $2N - p_m = p_k^s$; if $a_{m,k} = 1$, every row satisfies $R_i \ge s + 1$, forcing $\rho(M) \ge s + 1$ and $\mathcal{G}_{\mathbf{u}}(I) < (2N)^{\frac{1}{s+1}}$.

Proof of Proposition 4.18. 1. **Proof of (1):** For $i \in \{1, \dots, k-1\} \setminus \{m\}$, $a_{i,k} = 0$ by Lemma 3.9(2), so every prime factor of $2N - p_i$ belongs to $\{p_1, \dots, p_{k-1}\}$ and satisfies $p_j \le p_{k-1} \le \sqrt{2N} - 2 < \sqrt{2N - p_i}$; for $i = k$, $a_{k,k} = 0$ and $p_{k-1} \le \sqrt{2N} - 2 < \sqrt{\frac{2}{3}(2N)} < \sqrt{2N - p_k}$. Thus Proposition 4.5 gives $R_i \ge 3$ for all $i \ne m$. If $p_k \le \sqrt{2N}$ and $a_{m,k} = 1$ with $R_m = 2$, then $2N - p_m = p_k p_j$ for some $j \le k - 1$, which gives:

$$2N - \sqrt{2N} < 2N - p_m = p_k p_j \le \sqrt{2N}(\sqrt{2N} - 2) = 2N - 2\sqrt{2N}, \tag{47}$$

an immediate contradiction. Thus $R_m \ge 3$ as well, so $R_i \ge 3$ for all $i \in \{1, \dots, k\}$.

2. **Proof of (2):** When $R_i \ge 3$ for all i, Theorem 3.7 (6) gives $\rho(M) \ge \min_i R_i \ge 3$, and Theorem 4.4(1) yields (46). Similarly, if $p_k \le x := (2N)^{1/s}$ ($s \ge 2$) and $a_{m,k} = 1$, then any product of at most s prime factors from I containing at most one factor of p_k is bounded above by:

$$(p_k - 2)^{s-1} p_k \le (x-2)^{s-1} x \le x^{s-2}(x-2)x = x^s - 2x^{s-1} = 2N - 2x^{s-1} < 2N - x \le 2N - p_i, \tag{48}$$

forcing $R_i \ge s+1$ for all $i \in \{1, \dots, k\}$, which elevates the spectral radius to $\rho(M) \ge s+1$ and compresses $\mathcal{G}_{\mathbf{u}}(I) < (2N)^{\frac{1}{s+1}}$.

□

5 Summary of Results, Open Obstacles, and Future Directions

5.1 Summary of Proven Results

Across Sections 2–4, we have established that any hypothetical Goldbach counterexample $2N > 4 \cdot 10^{18}$ generates a non-empty stationary fixed point $P_\infty^* = D(P_\infty^*) \subset \mathcal{P}_{\le \frac{2N-5}{3}}$, whose minimal terminal islands $I = \{p_1 < p_2 < \cdots < p_k\} \subseteq P_\infty^*$ ($k \ge 2$) are governed by irreducible integer exponent matrices $M = (a_{i,j}) \in \mathbb{Z}_{\ge 0}^{k \times k}$ with $\mathrm{Tr}(M) = 0$, row sums $R_i = \sum_{j=1}^k a_{i,j} \ge 2$, and spectral radius $\rho(M) \ge 2$. Every such island is subject to the following universal invariants:

- **Coprime Base & Sub-Square Squeeze (Theorem 4.3):** $\gcd(2N - p_1, 2N - p_2) = 1$ ($S_1 \cap S_2 = \emptyset$), $3 \le p_1 < p_2 < \sqrt{2N}$, and $0 < \Lambda_{1,2} < \frac{2}{\sqrt{2N}} < 10^{-9}$.
- **Descending In-Degree Rigidity (Lemma 3.9):** $\text{in-deg}(p_k) = 1$, and each prime p_r ($r \ge 3$) divides at most one of $\{2N - p_1, \dots, 2N - p_{r-1}\}$.
- **Perron Spectral Laws (Theorem 4.4):** $\mathcal{G}_{\mathbf{u}}(I) = \prod_{j=1}^k p_j^{u_j} < (2N)^{1/\rho(M)} \le \sqrt{2N}$, $u_k \le \frac{u_m}{\rho(M)}$, and exact cancellation of $\ln p_k$ from the left Perron identity whenever $a_{m,k} = 1$.

Table 1 summarizes all unconditional and conditional eliminations and structural reductions proved in Sections 2–4.

Dimension (k)	Configuration / Regime	Status	Reference & Mechanism
Low-Dimensional Regimes ($k \le 5$)			
$k = 0$	Empty Limit Set ($P^*_\infty = \emptyset$)	**Unconditionally Eliminated**	Proposition 2.14 (Bertrand non-emptiness & finite stabilization)
$k = 1$	Single-Prime Loop ($2N - p_1 = p_1^{a_{1,1}}$)	**Unconditionally Eliminated**	Proposition 4.9 ($\mathrm{Tr}(M) = 0$, Non-Divisibility Inheritance)
$k = 2$	All $2N \le 10^{24}$	**Unconditionally Eliminated**	Proposition 4.10(4a) (Exhaustive search; $2N = 2200$ not a counterexample)
$k = 2$	Exponents $(a_1, a_2) \in \{(2,3),(2,4),(2,5)\}$	**Unconditionally Eliminated**	Proposition 4.10(3) (Polynomial identity obstruction $\forall 2N$)
$k = 2$	Fixed Base Primes (p_1, p_2) with $a_2 \to \infty$	**Unconditionally Eliminated**	Proposition 4.10(4b) (Laurent linear forms in two logarithms)
$k = 2$	Unrestricted Exponents $(a_1 \ge 3, a_2 \ge 4)$	**Conditional on** *abc*	Proposition 4.10(4c), Proposition 4.8 ($p_1 p_2^2 < (2N)^{11/12}$)
$k \in \{3,4\}$	All Square-Free Topologies ($a_{i,j} \in \{0,1\}$)	**Unconditionally Eliminated**	Propositions 4.11(1) & 4.12(2) (Difference & modular collapse)
$k \in \{3,4,5\}$	Bounded-Height Regimes	**Unconditionally Eliminated**	Propositions 4.7, 4.11–4.15 (Matveev bounds on $\Lambda_{1,2}$)
$k \in \{3,4,5\}$	Root-Compressed High-Multiplicity Regimes	**Conditional on** *abc*	Propositions 4.8, 4.11–4.15 ($p_2 \prod_{j \in S_1 \cup S_2} p_j \le (2N)^{1-\delta}$)
Universal & High-Dimensional Regimes ($k \ge 4$)			
$k \ge 2$	Upper Base Regime ($p_2 > \sqrt{2N}$)	**Unconditionally Eliminated**	Theorem 4.3(1) (Sub-square base bound $p_1 < p_2 < \sqrt{2N}$)
$k \ge 4$	Uniformly Dense Square-Free ($\lvert S_i \rvert = k - 1$)	**Unconditionally Eliminated**	Proposition 4.12(1) (Symmetric cofactor difference forces $Q = 1$)
$k \ge 5$	Symmetric Square-Free Pairs ($2N - p_a = p_b Q$)	**Unconditionally Eliminated**	Proposition 4.12(3) (Identical cofactor difference forces $Q = 1$)
$k \ge 2$	Sub-Square Islands ($p_k \le \sqrt{2N}, a_{m,k} = 1$)	**Compressed** ($\mathcal{G}_{\mathbf{u}} < (2N)^{1/3}$)	Proposition 4.18 ($R_i \ge 3\,\forall i \implies \rho(M) \ge 3 \implies p_1 < (2N)^{1/3}$)
$k \ge 6$	Single-Parent Contractible with $\overline{R}_5 \le \lfloor \frac{\ln(2N-3)}{\ln 3} \rfloor$	**Bounded to** $k \le 97$	Proposition 4.16 (Schur contraction $\overline{R}_5 > \frac{2k}{5}$ in $R_i \ge 3$ regime)

Table 1: Summary of proven eliminations and structural reductions for minimal terminal islands $I \subseteq P^*_\infty$.

5.2 Remaining Obstacles

To convert the reductions of Table 1 into a complete unconditional proof of the binary Goldbach Conjecture, three specific mathematical obstacles remain to be resolved:

1. **Unconditional High-Multiplicity Closing ($k \ge 2$):** When both the primes $p_j \in I$ and the exponents $a_{i,j}$ grow with $2N$ above the Baker–Matveev logarithmic height cutoff of Proposition 4.7, our elimination in Propositions 4.8–4.15 relies on the Masser–Oesterlé *abc* conjecture. Replacing this conditional step requires an unconditional upper bound on $\Lambda_{1,2}$ (or a two-equation Diophantine obstruction using $p_1 + \prod_{j \in S_1} p_j^{a_{1,j}} = p_2 + \prod_{r \in S_2} p_r^{a_{2,r}} = 2N$ simultaneously) when the prime bases vary.

2. **Low-Multiplicity Multi-Prime Base Support ($k \ge 3$):** When the disjoint base supports S_1, S_2 involve $\lvert S_1 \cup S_2 \rvert \ge 3$ primes with small exponents ($a_{i,j} \in \{1,2\}$) and large primes, the product $p_2 \prod_{j \in S_1 \cup S_2} p_j$ is not compressed below $(2N)^{1-\delta}$ (so Proposition 4.8 does not apply), while Matveev's constant $C(n) = 1.4 \cdot 30^{n+3} n^{4.5}$ in Proposition 4.7 grows exponentially with $n = \lvert S_1 \cup S_2 \rvert$.

3. **Super-Square Primes and Contracted Cofactor Growth ($k \ge 6$):** When $p_k > \sqrt{2N}$ and the unique parent complement is a Chen-type semiprime $2N - p_m = p_k q$ ($q \in I$, $R_m = 2$), the minimum row sum of M remains 2 rather than 3, preventing a direct Collatz–Wielandt jump to $\rho(M) \ge 3$ in Proposition 4.18. Furthermore, during the inductive Schur contraction of Proposition 4.16, any parent row m that absorbs an eliminated prime has its left-hand side multiplied by Q_m ($2N(Q_m - 1) + p_m$), which can exceed $2N$ and allow $\overline{R}_5$ to exceed the uncontracted ceiling $\lfloor \frac{\ln(2N-3)}{\ln 3} \rfloor$.

5.3 Possible Approaches to Overcome These Obstacles

The structural identities established in Sections 3 and 4 suggest four concrete avenues to attack the obstacles above:

1. **Diophantine Reciprocal Congruences at p_k:** In the Chen-semiprime bottleneck $2N - p_m = p_k q$ ($q \in I$), substituting $p_k = \frac{2N - p_m}{q}$ into $2N - p_k = Q_k = \prod_{j=1}^{k-1} p_j^{a_{k,j}}$ yields the exact reciprocal identity (Theorem 4.4(3)):

$$qQ_k = 2N(q-1) + p_m. \tag{49}$$

   Reducing (49) modulo q forces $q \nmid Q_k$ (since $q \mid p_m \implies p_m = q$, contradicting $a_{m,m} = 0$), which unconditionally proves $a_{k,q} = 0$. Reducing (49) modulo p_m shows that $a_{k,m} \ge 1 \implies p_m \mid (q-1)$, forcing $a_{k,m} = 0$ whenever $p_m > \frac{q-1}{2}$ and compressing $p_k < \frac{2N}{2p_m + 1}$ otherwise.

2. **Schur Resolvent Walk-Capacity Bounds:** Partitioning $M = \begin{pmatrix} M_{11} & \mathbf{e}_m \\ \mathbf{r}_k^T & 0 \end{pmatrix}$ at node k gives the Schur resolvent expansion

$$\rho(M)^2 = \mathbf{r}_k^T(\rho(M)I_{k-1} - M_{11})^{-1}\mathbf{e}_m = a_{k,m} + \frac{a_{k,q}}{\rho(M)} + \sum_{\ell=2}^{\infty} \frac{\mathbf{r}_k^T M_{11}^{\ell}\mathbf{e}_m}{\rho(M)^{\ell}}. \tag{50}$$

Because $a_{k,q} = 0$ (and $a_{k,m} = 0$ when $p_m \nmid (q-1)$), short return walks through p_k vanish, creating an upper bound on $\rho(M)$ that competes directly against the Collatz–Wielandt lower bound when $R_i \geq 3$ for $i \neq m$.

3. **Sub-Alphabet Perron Cancellation:** Whenever $a_{m,k} = 1$, Theorem 4.4(4) eliminates $\ln p_k$ identically from the left Perron equation $\mathbf{u}^T\mathbf{y} = \rho(M)\mathbf{u}^T\mathbf{x}$, leaving an exact spectral balance on the $(k-1)$-prime sub-alphabet $\{p_1, \ldots, p_{k-1}\}$:

$$\sum_{i \neq m,k} u_i \ln(2N - p_i) + u_m \ln Q_m = \sum_{j=1}^{k-1} \left(\sum_{i=1}^{k-1} u_i a_{i,j} \right) \ln p_j. \tag{51}$$

Using this reduced identity recursively on super-square primes $p_\ell > \sqrt{2N}$ offers a pathway to compress the geometric mean of the sub-square primes without accumulating cofactor inflation.

4. **Coupled System Obstructions and In-Degree Sparsity:** Whereas Proposition 4.7 applies Matveev's theorem only to the single difference $(2N - p_1) - (2N - p_2) = p_2 - p_1$, the full system $\mathcal{S}(2N, k, M)$ imposes k simultaneous equations $p_i + \prod_{j=1}^{k} p_j^{a_{i,j}} = 2N$ coupled by the Descending In-Degree Cascade of Lemma 3.9(3) (which restricts each p_r to at most one of $\{2N - p_1, \ldots, 2N - p_{r-1}\}$). Exploiting these k coupled equations simultaneously via multi-linear transcendence bounds or p-adic interpolation determinants [5] is a promising route to eliminate the remaining regimes unconditionally.

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